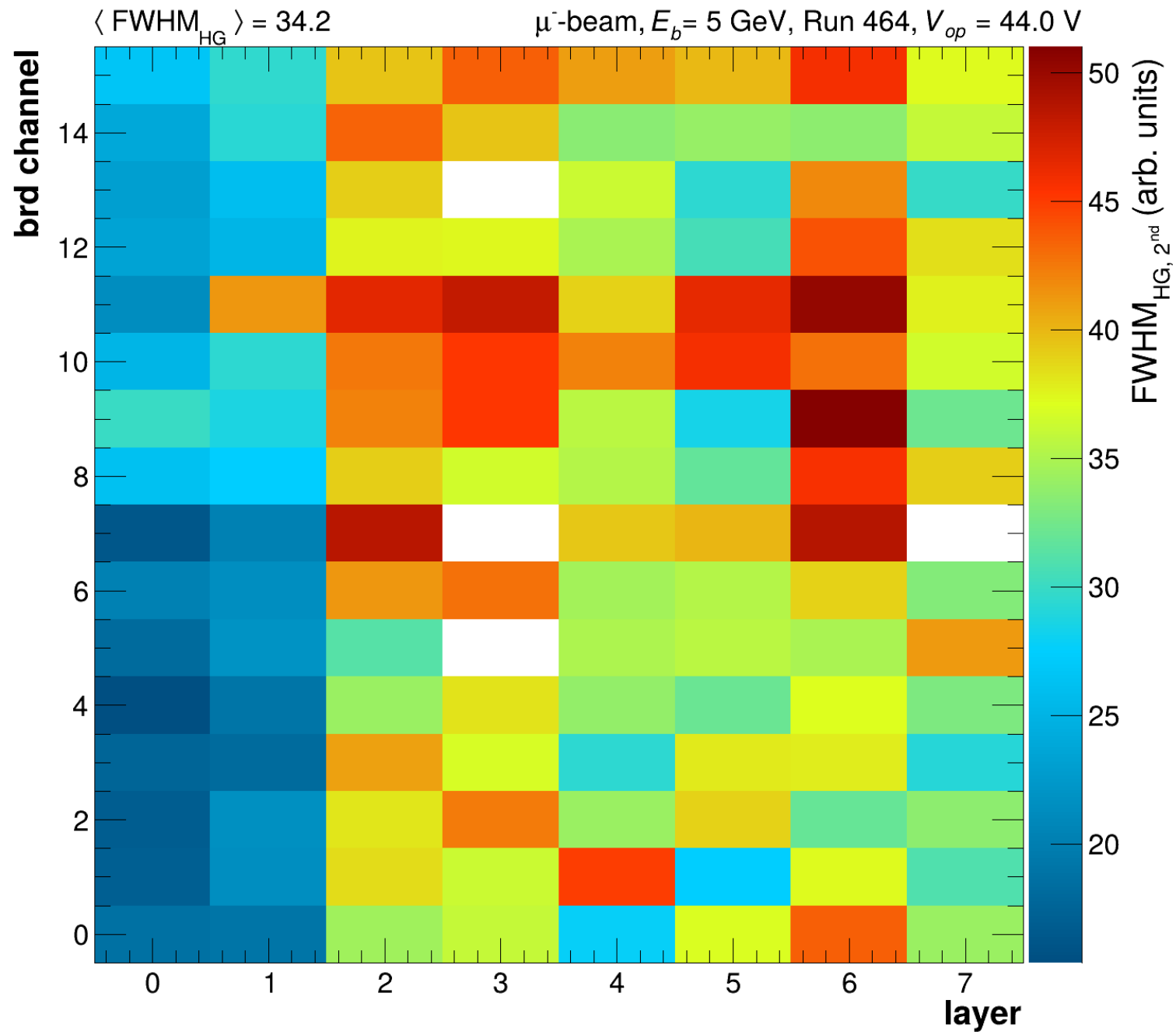
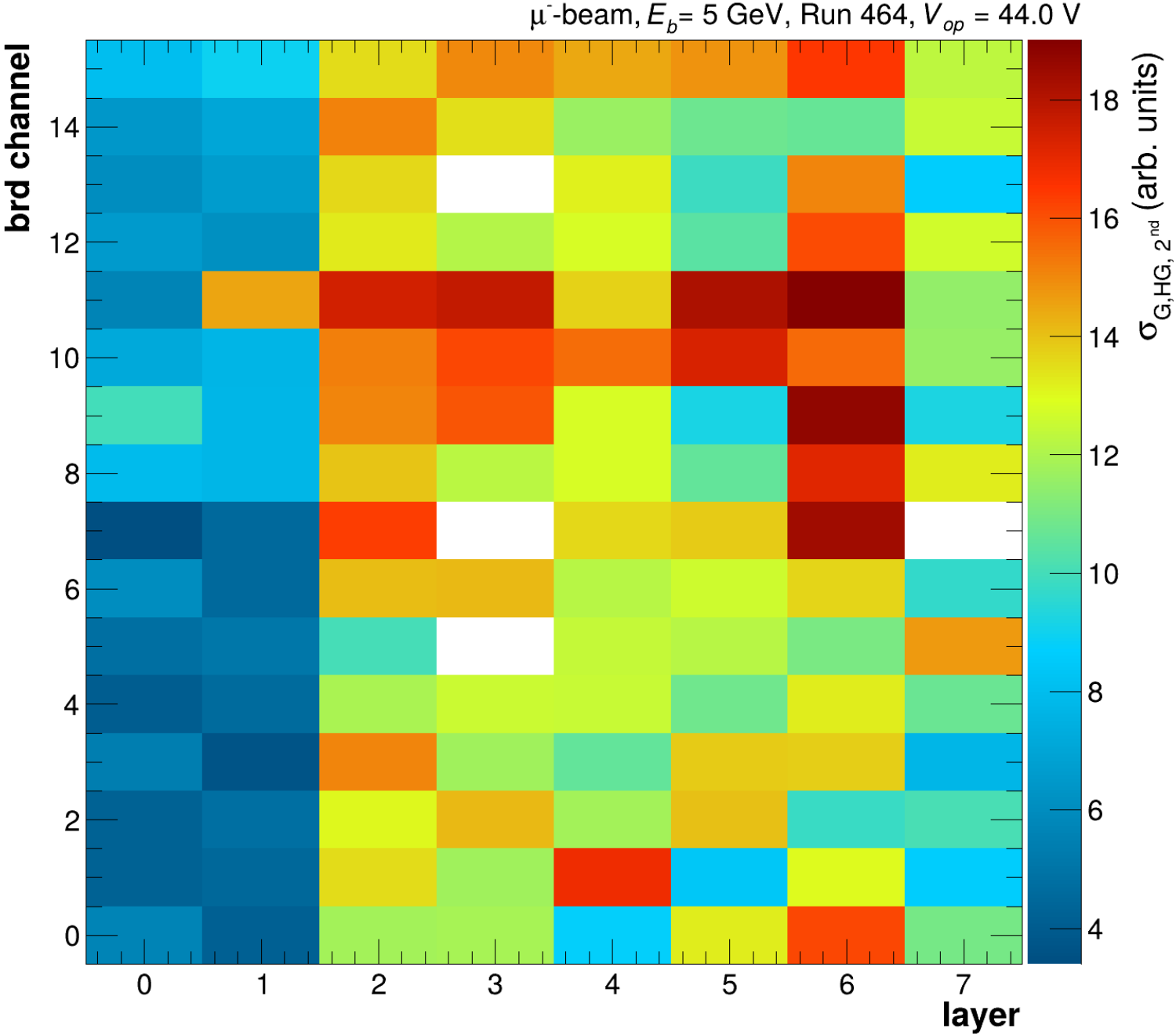
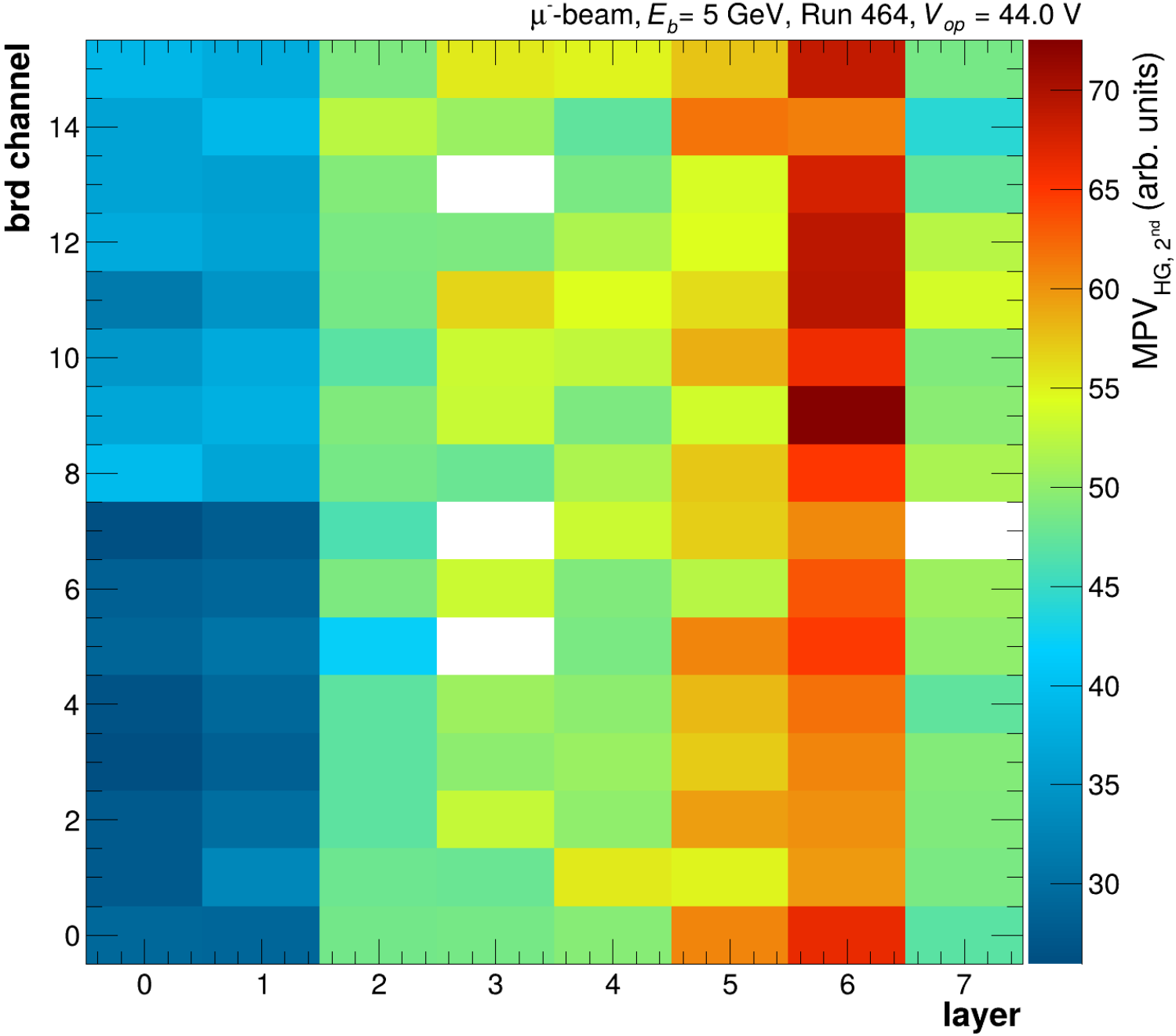
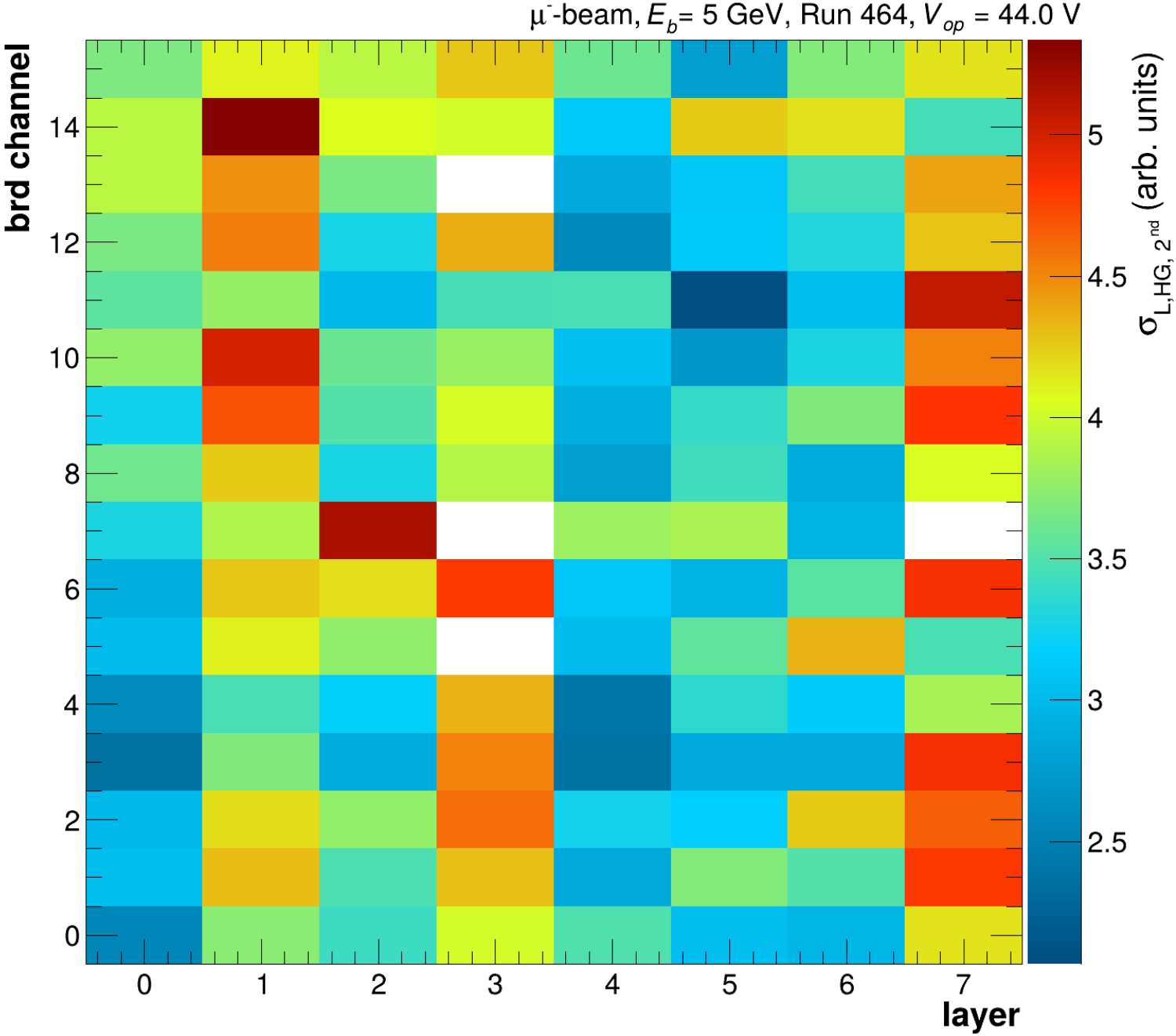
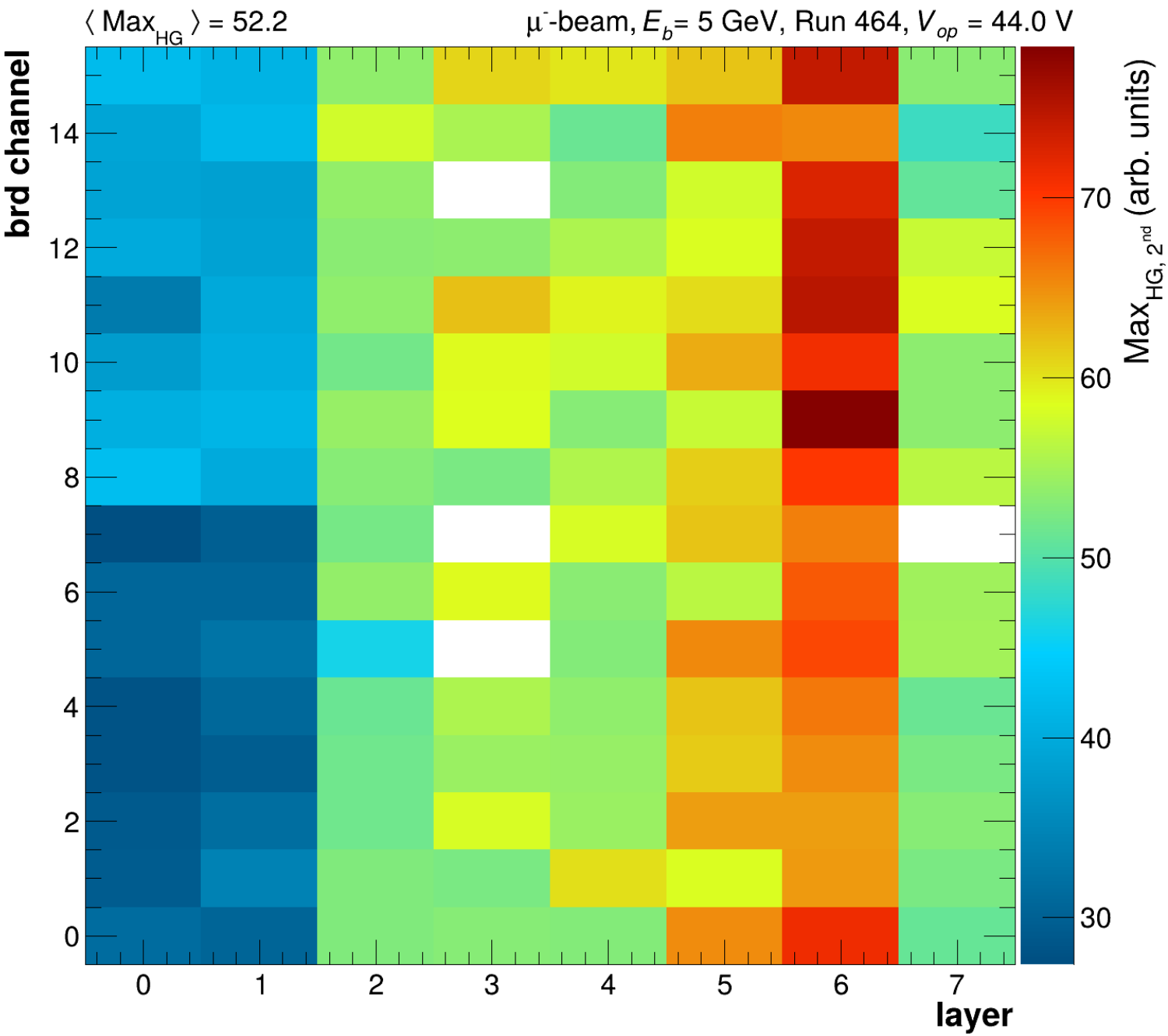
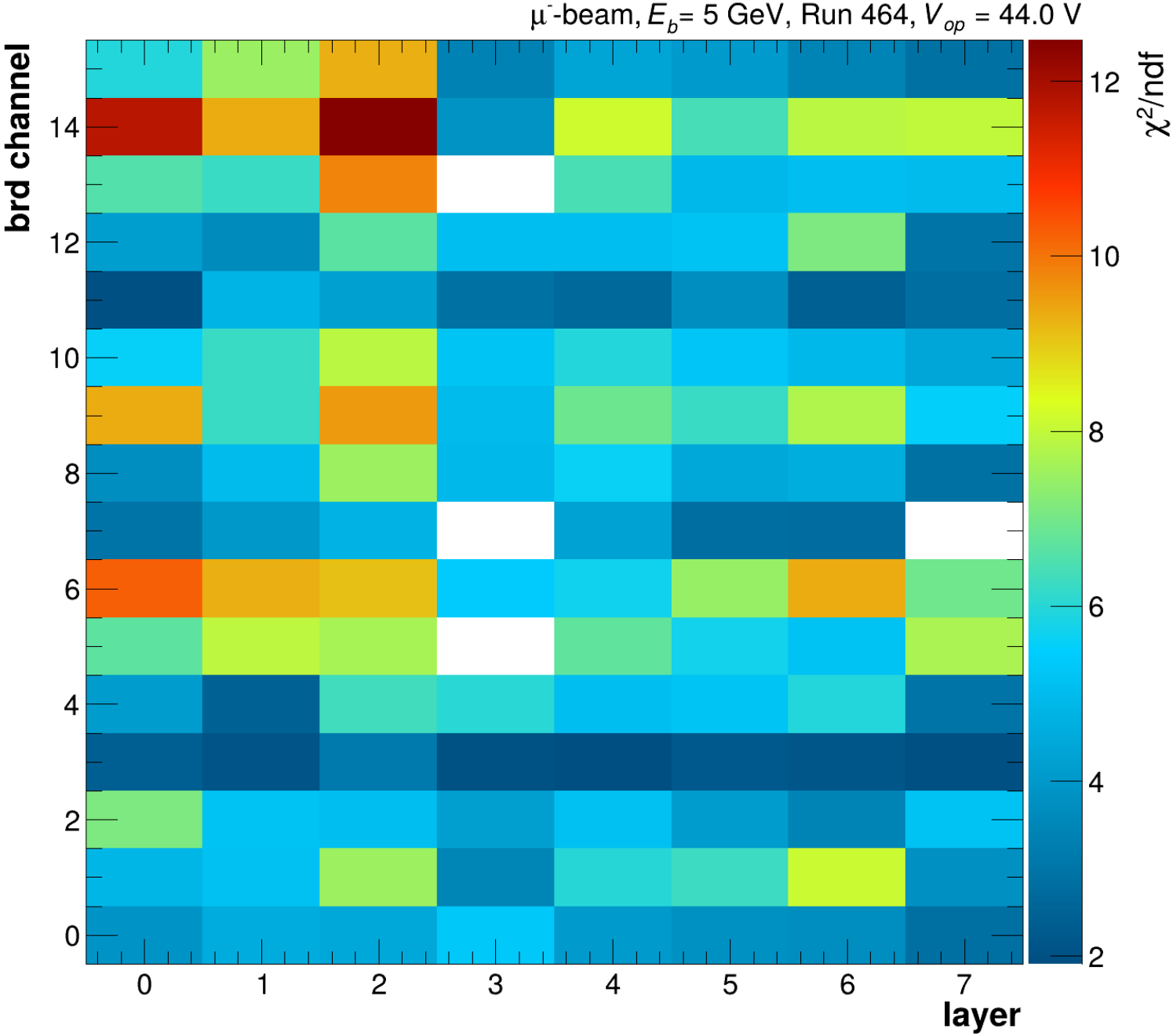

⟨ FWHM_HG ⟩ = 34.2
μ⁻-beam, E_b = 5 GeV, Run 464, V_op = 44.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
25
30
35
40
45
50

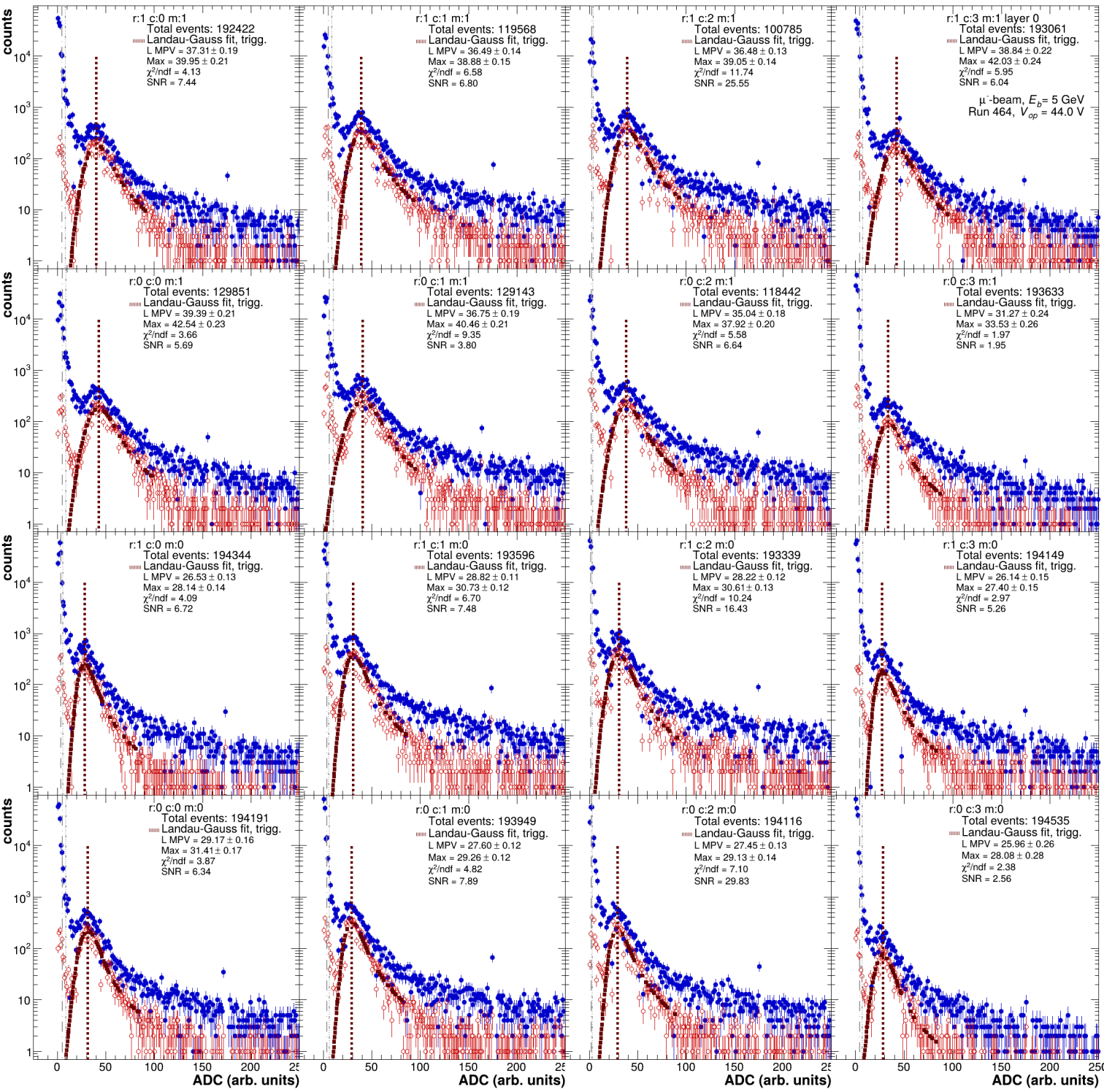

μ-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
4
6
8
10
12
14
16
18

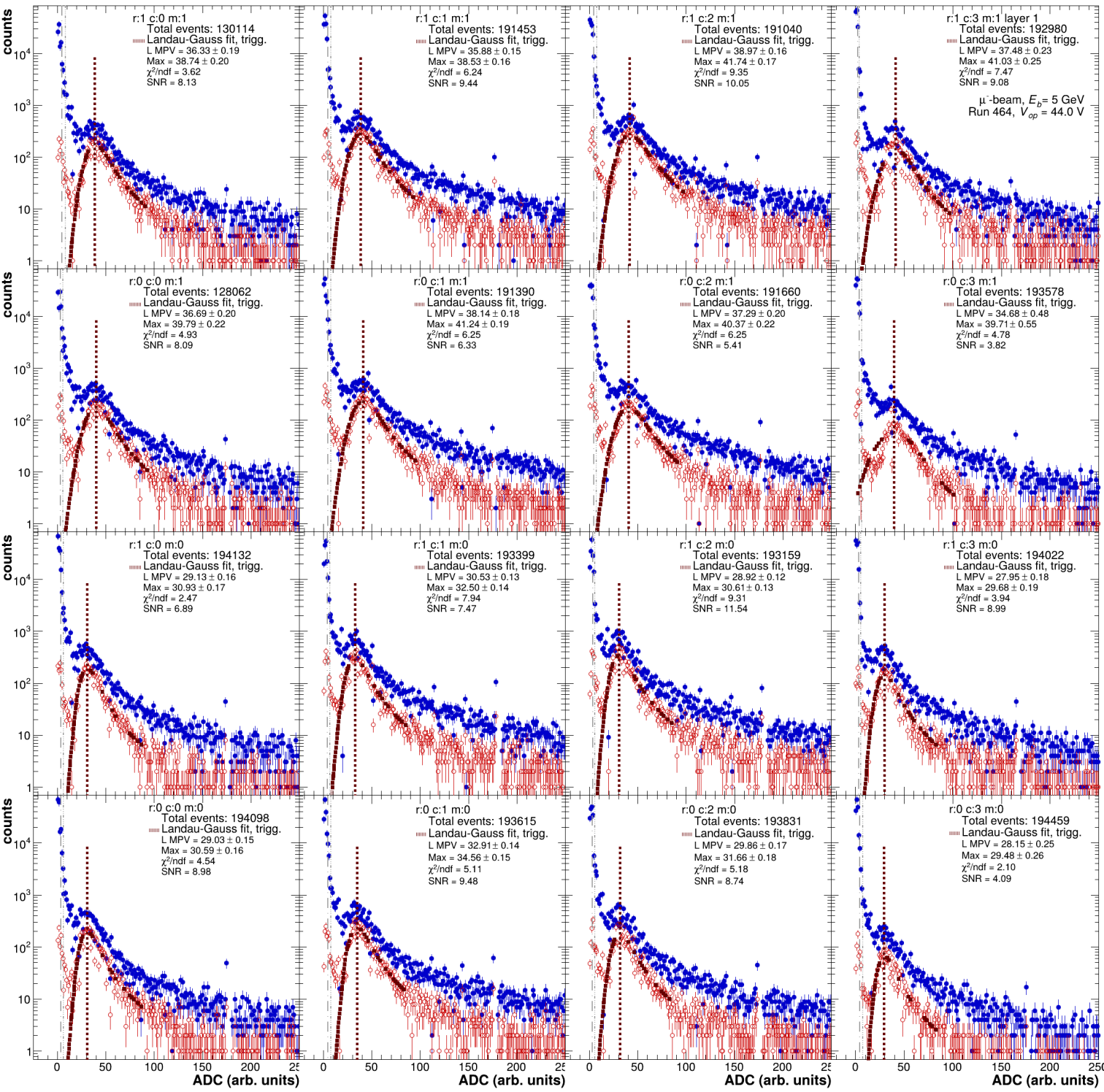

μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
30
35
40
45
50
55
60
65
70

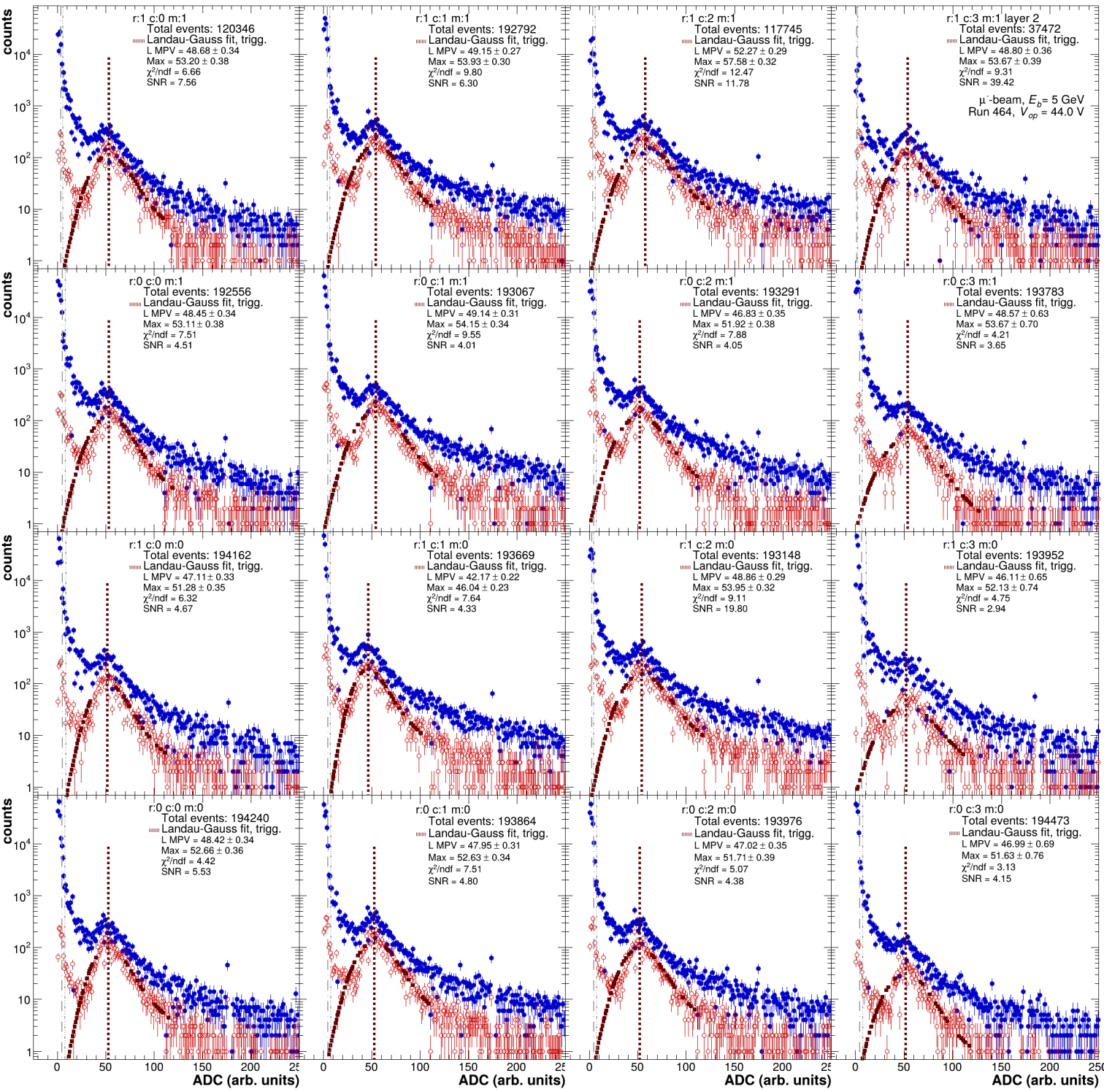

μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
2.5
3
3.5
4
4.5
5

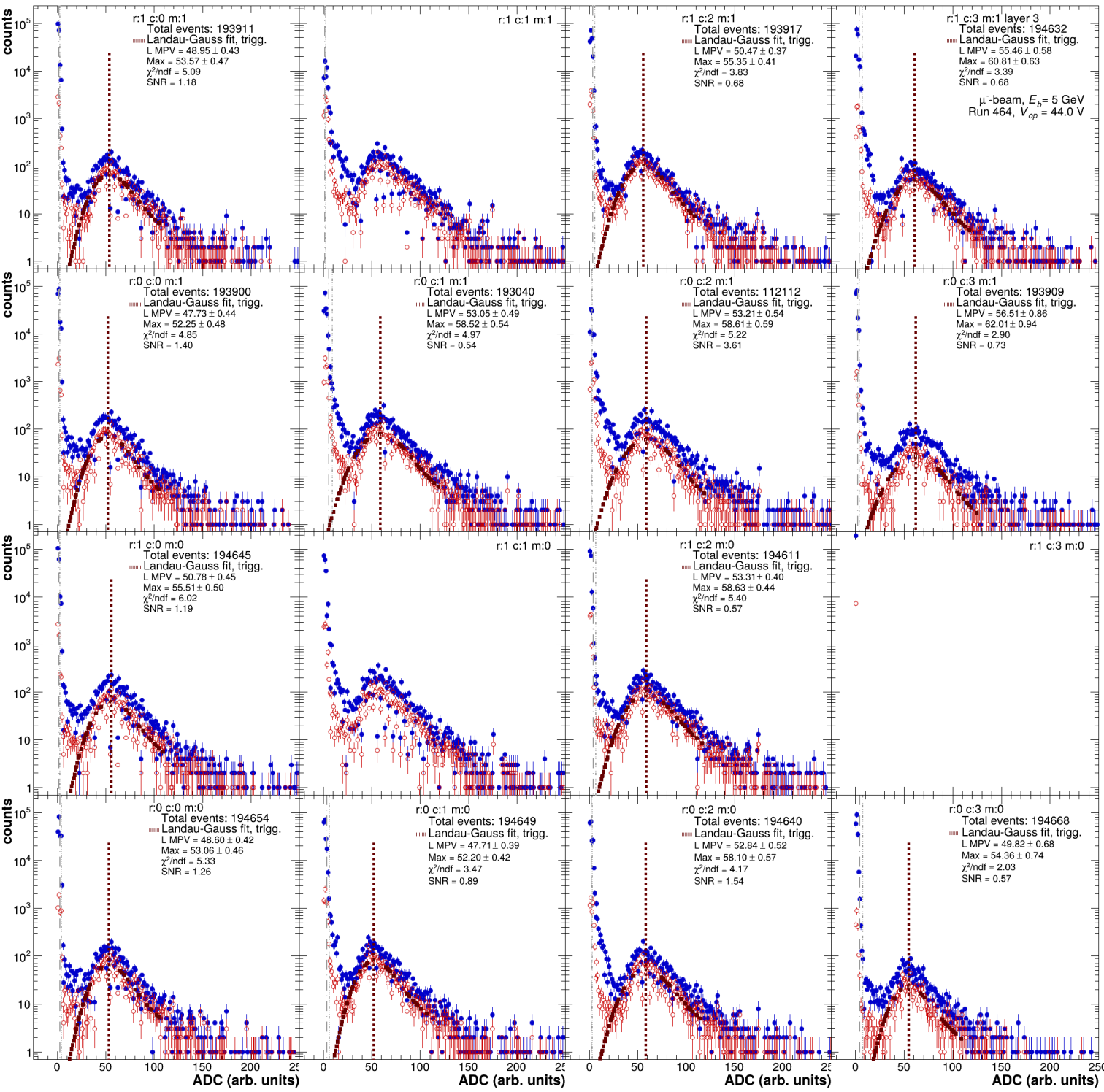

⟨ Max_HG ⟩ = 52.2
μ⁻-beam, E_b = 5 GeV, Run 464, V_op = 44.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
40
50
60
70

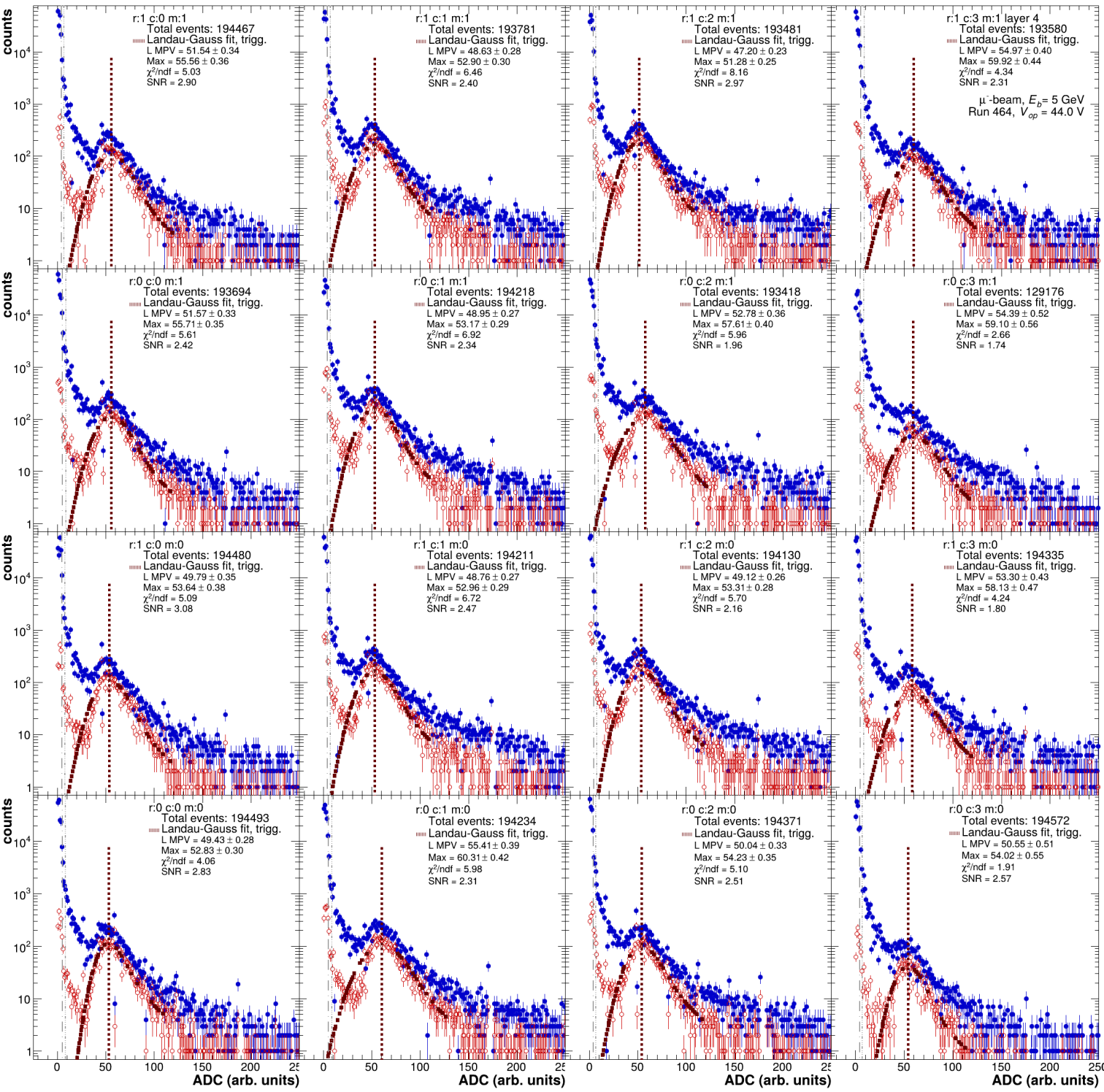

μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
layer
χ^2/ndf
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
2
4
6
8
10
12

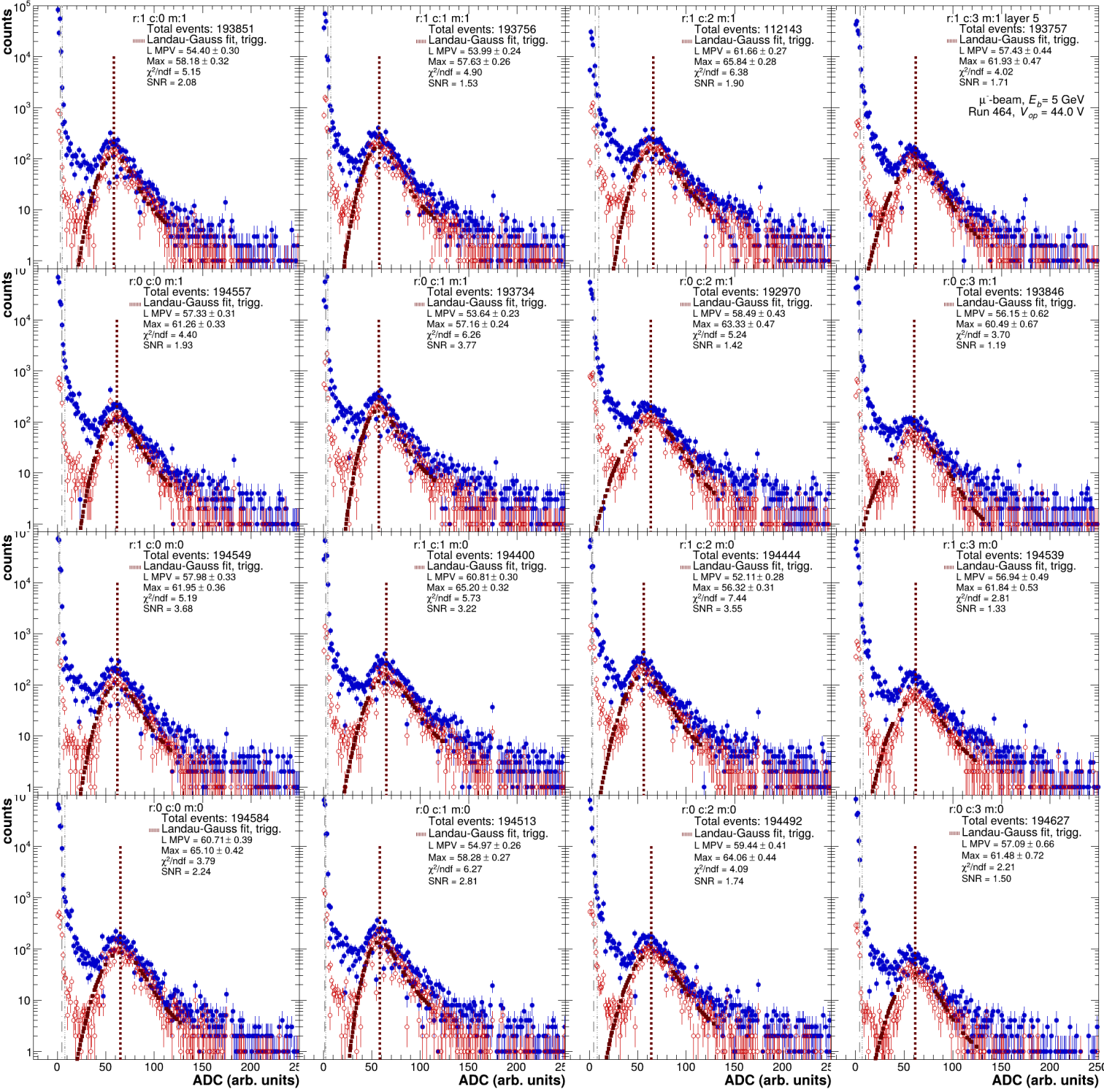

r:1 c:0 m:1
Total events: 192422
Landau-Gauss fit, trigg.
L MPV = 37.31 ± 0.19
Max = 39.95 ± 0.21
χ²/ndf = 4.13
SNR = 7.44
r:1 c:1 m:1
Total events: 119568
Landau-Gauss fit, trigg.
L MPV = 36.49 ± 0.14
Max = 38.88 ± 0.15
χ²/ndf = 6.58
SNR = 6.80
r:1 c:2 m:1
Total events: 100785
Landau-Gauss fit, trigg.
L MPV = 36.48 ± 0.13
Max = 39.05 ± 0.14
χ²/ndf = 11.74
SNR = 25.55
r:1 c:3 m:1 layer 0
Total events: 193061
Landau-Gauss fit, trigg.
L MPV = 38.84 ± 0.22
Max = 42.03 ± 0.24
χ²/ndf = 5.95
SNR = 6.04
μ⁻-beam, E_b = 5 GeV
Run 464, V_op = 44.0 V
r:0 c:0 m:1
Total events: 129851
Landau-Gauss fit, trigg.
L MPV = 39.39 ± 0.21
Max = 42.54 ± 0.23
χ²/ndf = 3.66
SNR = 5.69
r:0 c:1 m:1
Total events: 129143
Landau-Gauss fit, trigg.
L MPV = 36.75 ± 0.19
Max = 40.46 ± 0.21
χ²/ndf = 9.35
SNR = 3.80
r:0 c:2 m:1
Total events: 118442
Landau-Gauss fit, trigg.
L MPV = 35.04 ± 0.18
Max = 37.92 ± 0.20
χ²/ndf = 5.58
SNR = 6.64
r:0 c:3 m:1
Total events: 193633
Landau-Gauss fit, trigg.
L MPV = 31.27 ± 0.24
Max = 33.53 ± 0.26
χ²/ndf = 1.97
SNR = 1.95
r:1 c:0 m:0
Total events: 194344
Landau-Gauss fit, trigg.
L MPV = 26.53 ± 0.13
Max = 28.14 ± 0.14
χ²/ndf = 4.09
SNR = 6.72
r:1 c:1 m:0
Total events: 193596
Landau-Gauss fit, trigg.
L MPV = 28.82 ± 0.11
Max = 30.73 ± 0.12
χ²/ndf = 6.70
SNR = 7.48
r:1 c:2 m:0
Total events: 193339
Landau-Gauss fit, trigg.
L MPV = 28.22 ± 0.12
Max = 30.61 ± 0.13
χ²/ndf = 10.24
SNR = 16.43
r:1 c:3 m:0
Total events: 194149
Landau-Gauss fit, trigg.
L MPV = 26.14 ± 0.15
Max = 27.40 ± 0.15
χ²/ndf = 2.97
SNR = 5.26
r:0 c:0 m:0
Total events: 194191
Landau-Gauss fit, trigg.
L MPV = 29.17 ± 0.16
Max = 31.41 ± 0.17
χ²/ndf = 3.87
SNR = 6.34
r:0 c:1 m:0
Total events: 193949
Landau-Gauss fit, trigg.
L MPV = 27.60 ± 0.12
Max = 29.26 ± 0.12
χ²/ndf = 4.82
SNR = 7.89
r:0 c:2 m:0
Total events: 194116
Landau-Gauss fit, trigg.
L MPV = 27.45 ± 0.13
Max = 29.13 ± 0.14
χ²/ndf = 7.10
SNR = 29.83
r:0 c:3 m:0
Total events: 194535
Landau-Gauss fit, trigg.
L MPV = 25.96 ± 0.26
Max = 28.08 ± 0.28
χ²/ndf = 2.38
SNR = 2.56
counts
ADC (arb. units)

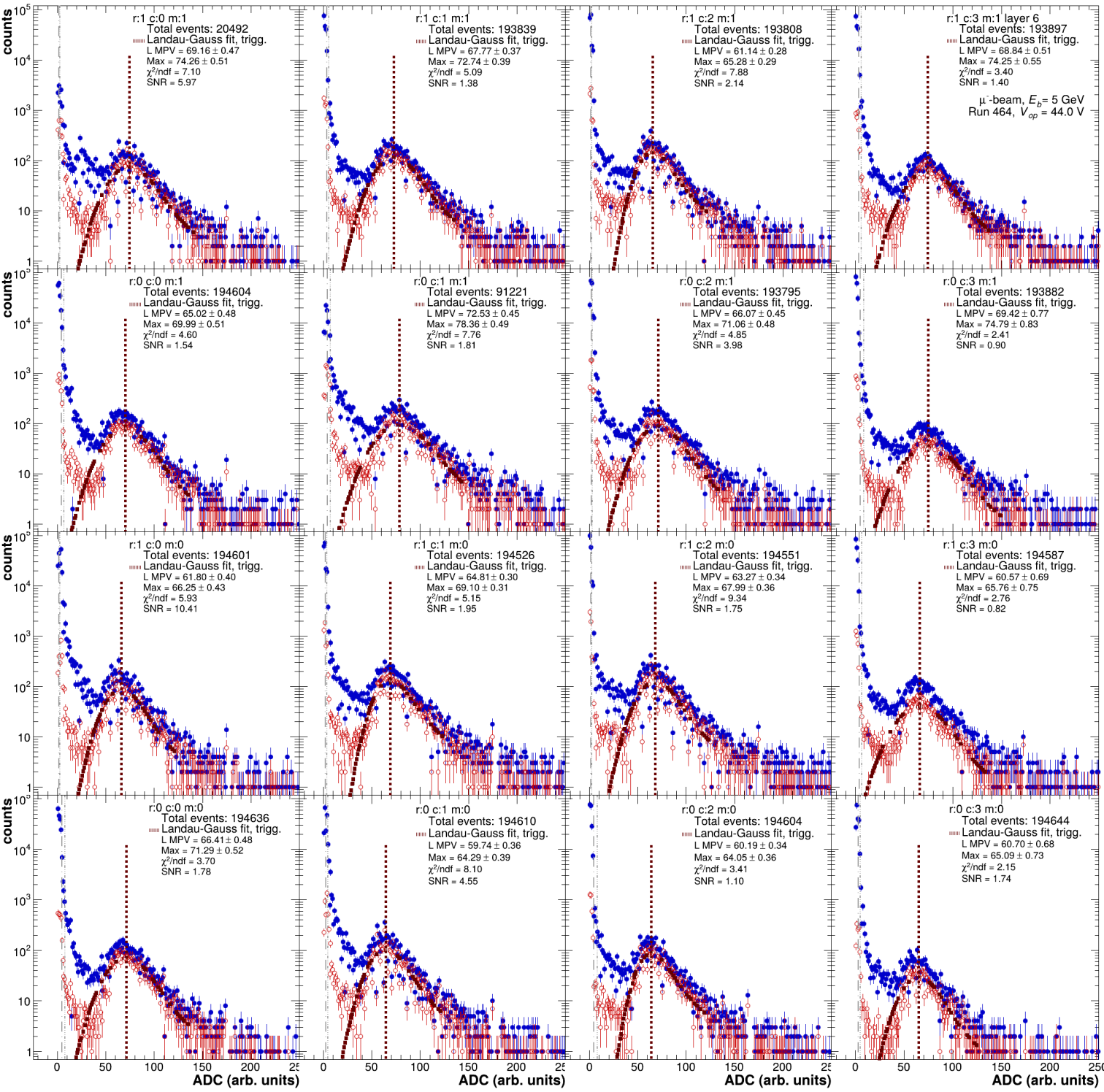

r:1 c:0 m:1
Total events: 130114
Landau-Gauss fit, trigg.
L MPV = 36.33 ± 0.19
Max = 38.74 ± 0.20
χ²/ndf = 3.62
SNR = 8.13
r:1 c:1 m:1
Total events: 191453
Landau-Gauss fit, trigg.
L MPV = 35.88 ± 0.15
Max = 38.53 ± 0.16
χ²/ndf = 6.24
SNR = 9.44
r:1 c:2 m:1
Total events: 191040
Landau-Gauss fit, trigg.
L MPV = 38.97 ± 0.16
Max = 41.74 ± 0.17
χ²/ndf = 9.35
SNR = 10.05
r:1 c:3 m:1 layer 1
Total events: 192980
Landau-Gauss fit, trigg.
L MPV = 37.48 ± 0.23
Max = 41.03 ± 0.25
χ²/ndf = 7.47
SNR = 9.08
μ⁻-beam, E_b = 5 GeV
Run 464, V_op = 44.0 V
r:0 c:0 m:1
Total events: 128062
Landau-Gauss fit, trigg.
L MPV = 36.69 ± 0.20
Max = 39.79 ± 0.22
χ²/ndf = 4.93
SNR = 8.09
r:0 c:1 m:1
Total events: 191390
Landau-Gauss fit, trigg.
L MPV = 38.14 ± 0.18
Max = 41.24 ± 0.19
χ²/ndf = 6.25
SNR = 6.33
r:0 c:2 m:1
Total events: 191660
Landau-Gauss fit, trigg.
L MPV = 37.29 ± 0.20
Max = 40.37 ± 0.22
χ²/ndf = 6.25
SNR = 5.41
r:0 c:3 m:1
Total events: 193578
Landau-Gauss fit, trigg.
L MPV = 34.68 ± 0.48
Max = 39.71 ± 0.55
χ²/ndf = 4.78
SNR = 3.82
r:1 c:0 m:0
Total events: 194132
Landau-Gauss fit, trigg.
L MPV = 29.13 ± 0.16
Max = 30.93 ± 0.17
χ²/ndf = 2.47
SNR = 6.89
r:1 c:1 m:0
Total events: 193399
Landau-Gauss fit, trigg.
L MPV = 30.53 ± 0.13
Max = 32.50 ± 0.14
χ²/ndf = 7.94
SNR = 7.47
r:1 c:2 m:0
Total events: 193159
Landau-Gauss fit, trigg.
L MPV = 28.92 ± 0.12
Max = 30.61 ± 0.13
χ²/ndf = 9.31
SNR = 11.54
r:1 c:3 m:0
Total events: 194022
Landau-Gauss fit, trigg.
L MPV = 27.95 ± 0.18
Max = 29.68 ± 0.19
χ²/ndf = 3.94
SNR = 8.99
r:0 c:0 m:0
Total events: 194098
Landau-Gauss fit, trigg.
L MPV = 29.03 ± 0.15
Max = 30.59 ± 0.16
χ²/ndf = 4.54
SNR = 8.98
r:0 c:1 m:0
Total events: 193615
Landau-Gauss fit, trigg.
L MPV = 32.91 ± 0.14
Max = 34.56 ± 0.15
χ²/ndf = 5.11
SNR = 9.48
r:0 c:2 m:0
Total events: 193831
Landau-Gauss fit, trigg.
L MPV = 29.86 ± 0.17
Max = 31.66 ± 0.18
χ²/ndf = 5.18
SNR = 8.74
r:0 c:3 m:0
Total events: 194459
Landau-Gauss fit, trigg.
L MPV = 28.15 ± 0.25
Max = 29.48 ± 0.26
χ²/ndf = 2.10
SNR = 4.09
counts
ADC (arb. units)

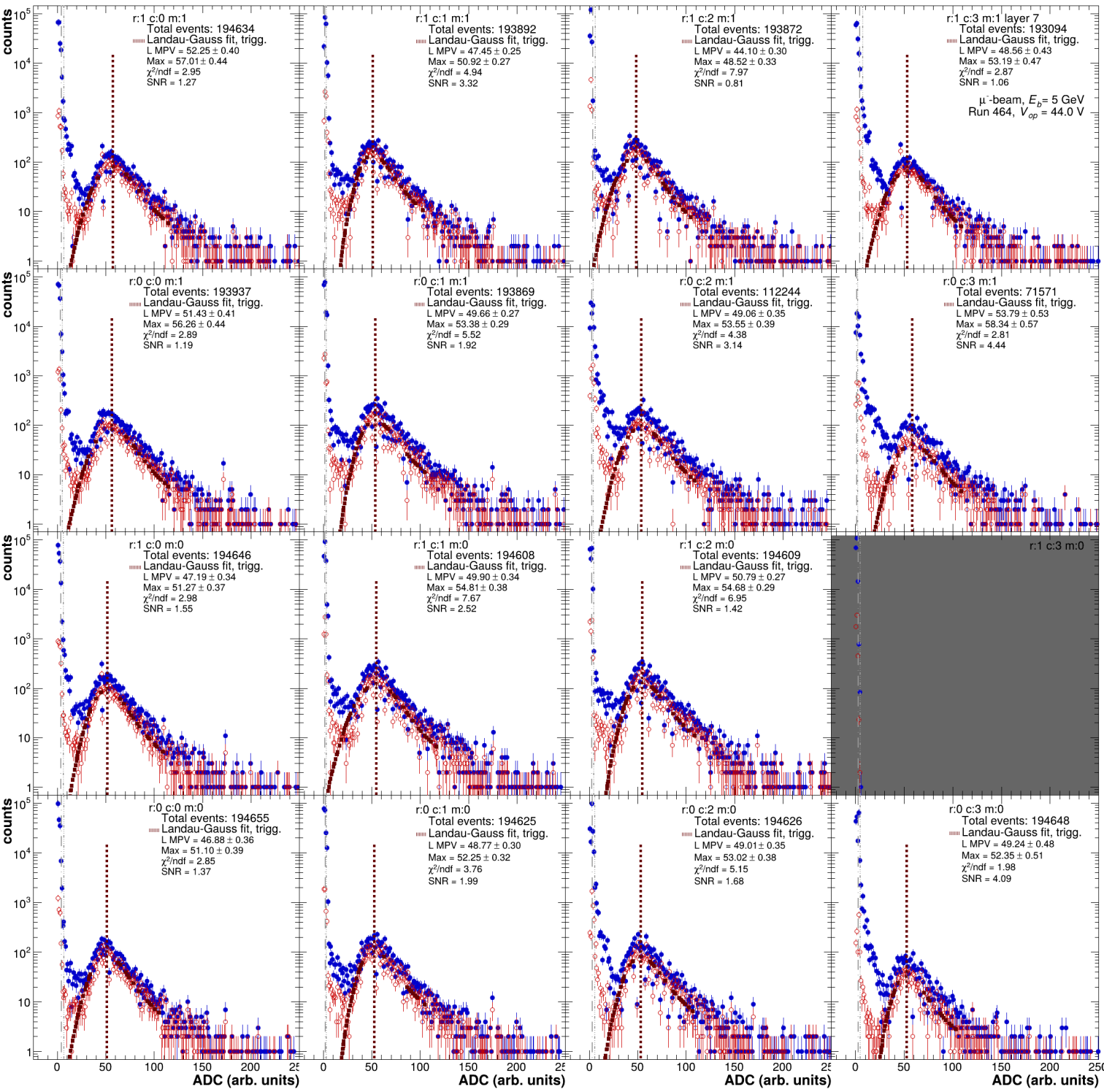

r:1 c:0 m:1
Total events: 120346
Landau-Gauss fit, trigg.
L MPV = 48.68 ± 0.34
Max = 53.20 ± 0.38
χ²/ndf = 6.66
SNR = 7.56
r:1 c:1 m:1
Total events: 192792
Landau-Gauss fit, trigg.
L MPV = 49.15 ± 0.27
Max = 53.93 ± 0.30
χ²/ndf = 9.80
SNR = 6.30
r:1 c:2 m:1
Total events: 117745
Landau-Gauss fit, trigg.
L MPV = 52.27 ± 0.29
Max = 57.58 ± 0.32
χ²/ndf = 12.47
SNR = 11.78
r:1 c:3 m:1 layer 2
Total events: 37472
Landau-Gauss fit, trigg.
L MPV = 48.80 ± 0.36
Max = 53.67 ± 0.39
χ²/ndf = 9.31
SNR = 39.42
μ⁻-beam, E_b = 5 GeV
Run 464, V_op = 44.0 V
r:0 c:0 m:1
Total events: 192556
Landau-Gauss fit, trigg.
L MPV = 48.45 ± 0.34
Max = 53.11 ± 0.38
χ²/ndf = 7.51
SNR = 4.51
r:0 c:1 m:1
Total events: 193067
Landau-Gauss fit, trigg.
L MPV = 49.14 ± 0.31
Max = 54.15 ± 0.34
χ²/ndf = 9.55
SNR = 4.01
r:0 c:2 m:1
Total events: 193291
Landau-Gauss fit, trigg.
L MPV = 46.83 ± 0.35
Max = 51.92 ± 0.38
χ²/ndf = 7.88
SNR = 4.05
r:0 c:3 m:1
Total events: 193783
Landau-Gauss fit, trigg.
L MPV = 48.57 ± 0.63
Max = 53.67 ± 0.70
χ²/ndf = 4.21
SNR = 3.65
r:1 c:0 m:0
Total events: 194162
Landau-Gauss fit, trigg.
L MPV = 47.11 ± 0.33
Max = 51.28 ± 0.35
χ²/ndf = 6.32
SNR = 4.67
r:1 c:1 m:0
Total events: 193669
Landau-Gauss fit, trigg.
L MPV = 42.17 ± 0.22
Max = 46.04 ± 0.23
χ²/ndf = 7.64
SNR = 4.33
r:1 c:2 m:0
Total events: 193148
Landau-Gauss fit, trigg.
L MPV = 48.86 ± 0.29
Max = 53.95 ± 0.32
χ²/ndf = 9.11
SNR = 19.80
r:1 c:3 m:0
Total events: 193952
Landau-Gauss fit, trigg.
L MPV = 46.11 ± 0.65
Max = 52.13 ± 0.74
χ²/ndf = 4.75
SNR = 2.94
r:0 c:0 m:0
Total events: 194240
Landau-Gauss fit, trigg.
L MPV = 48.42 ± 0.34
Max = 52.66 ± 0.36
χ²/ndf = 4.42
SNR = 5.53
r:0 c:1 m:0
Total events: 193864
Landau-Gauss fit, trigg.
L MPV = 47.95 ± 0.31
Max = 52.63 ± 0.34
χ²/ndf = 7.51
SNR = 4.80
r:0 c:2 m:0
Total events: 193976
Landau-Gauss fit, trigg.
L MPV = 47.02 ± 0.35
Max = 51.71 ± 0.39
χ²/ndf = 5.07
SNR = 4.38
r:0 c:3 m:0
Total events: 194473
Landau-Gauss fit, trigg.
L MPV = 46.99 ± 0.69
Max = 51.63 ± 0.76
χ²/ndf = 3.13
SNR = 4.15
counts
ADC (arb. units)

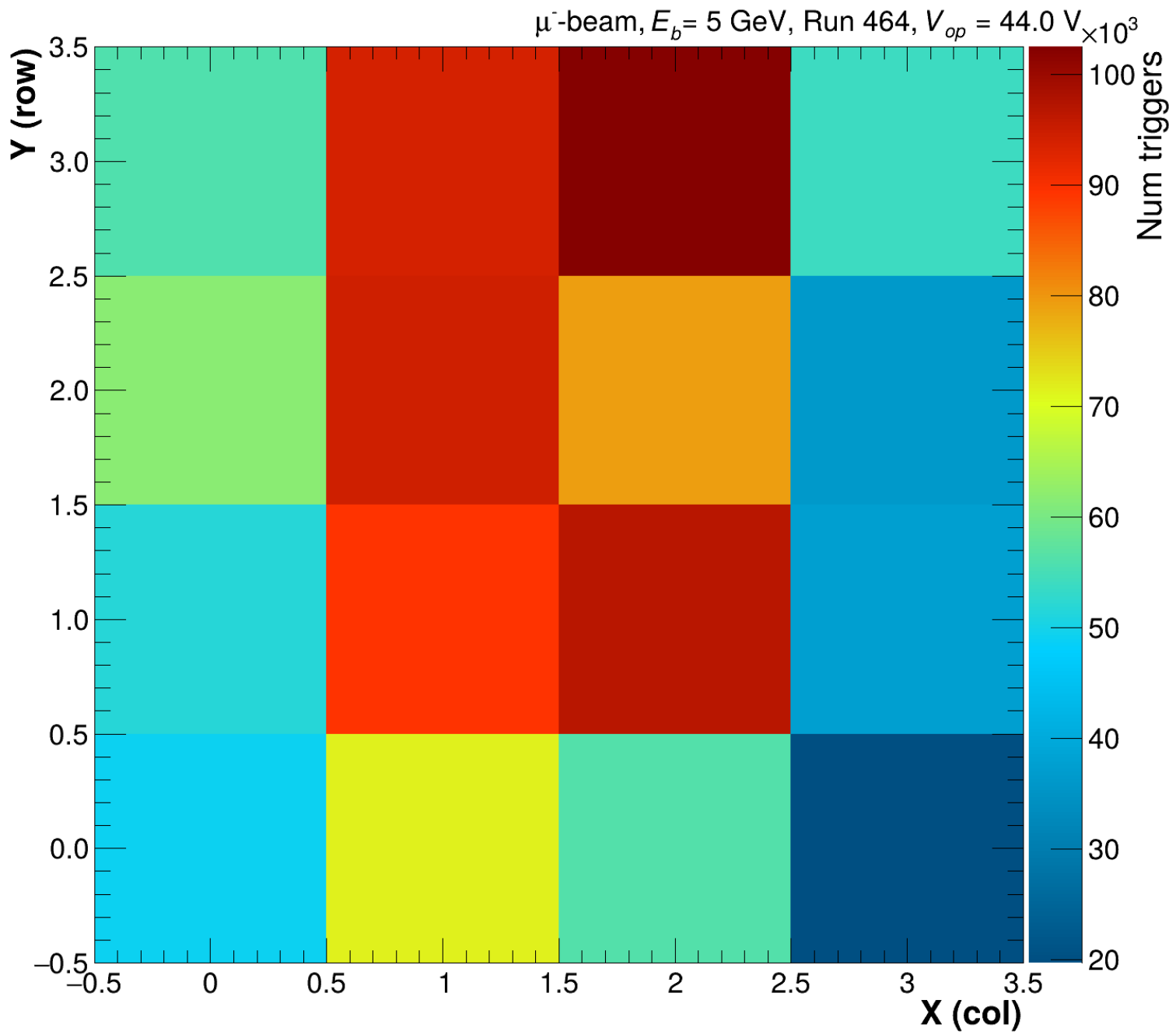

r:1 c:0 m:1
Total events: 193911
Landau-Gauss fit, trigg.
L MPV = 48.95 ± 0.43
Max = 53.57 ± 0.47
χ²/ndf = 5.09
SNR = 1.18
r:1 c:1 m:1
r:1 c:2 m:1
Total events: 193917
Landau-Gauss fit, trigg.
L MPV = 50.47 ± 0.37
Max = 55.35 ± 0.41
χ²/ndf = 3.83
SNR = 0.68
r:1 c:3 m:1 layer 3
Total events: 194632
Landau-Gauss fit, trigg.
L MPV = 55.46 ± 0.58
Max = 60.81 ± 0.63
χ²/ndf = 3.39
SNR = 0.68
μ-beam, E_b = 5 GeV
Run 464, V_op = 44.0 V
r:0 c:0 m:1
Total events: 193900
Landau-Gauss fit, trigg.
L MPV = 47.73 ± 0.44
Max = 52.25 ± 0.48
χ²/ndf = 4.85
SNR = 1.40
r:0 c:1 m:1
Total events: 193040
Landau-Gauss fit, trigg.
L MPV = 53.05 ± 0.49
Max = 58.52 ± 0.54
χ²/ndf = 4.97
SNR = 0.54
r:0 c:2 m:1
Total events: 112112
Landau-Gauss fit, trigg.
L MPV = 53.21 ± 0.54
Max = 58.61 ± 0.59
χ²/ndf = 5.22
SNR = 3.61
r:0 c:3 m:1
Total events: 193909
Landau-Gauss fit, trigg.
L MPV = 56.51 ± 0.86
Max = 62.01 ± 0.94
χ²/ndf = 2.90
SNR = 0.73
r:1 c:0 m:0
Total events: 194645
Landau-Gauss fit, trigg.
L MPV = 50.78 ± 0.45
Max = 55.51 ± 0.50
χ²/ndf = 6.02
SNR = 1.19
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 194611
Landau-Gauss fit, trigg.
L MPV = 53.31 ± 0.40
Max = 58.63 ± 0.44
χ²/ndf = 5.40
SNR = 0.57
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 194654
Landau-Gauss fit, trigg.
L MPV = 48.60 ± 0.42
Max = 53.06 ± 0.46
χ²/ndf = 5.33
SNR = 1.26
r:0 c:1 m:0
Total events: 194649
Landau-Gauss fit, trigg.
L MPV = 47.71 ± 0.39
Max = 52.20 ± 0.42
χ²/ndf = 3.47
SNR = 0.89
r:0 c:2 m:0
Total events: 194640
Landau-Gauss fit, trigg.
L MPV = 52.84 ± 0.52
Max = 58.10 ± 0.57
χ²/ndf = 4.17
SNR = 1.54
r:0 c:3 m:0
Total events: 194668
Landau-Gauss fit, trigg.
L MPV = 49.82 ± 0.68
Max = 54.36 ± 0.74
χ²/ndf = 2.03
SNR = 0.57
counts
ADC (arb. units)

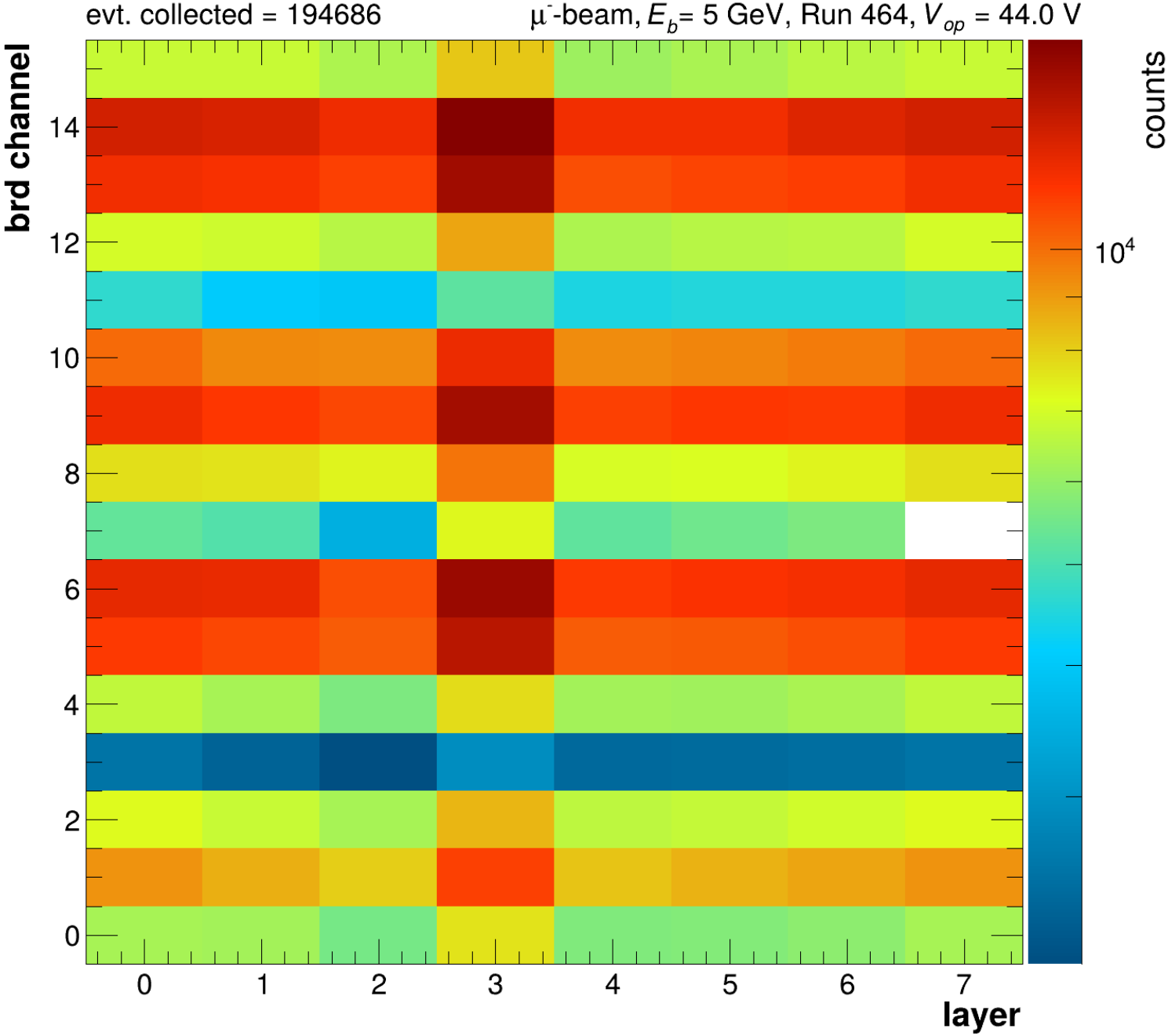

r:1 c:0 m:1
Total events: 194467
Landau-Gauss fit, trigg.
L MPV = 51.54 ± 0.34
Max = 55.56 ± 0.36
χ²/ndf = 5.03
SNR = 2.90
r:1 c:1 m:1
Total events: 193781
Landau-Gauss fit, trigg.
L MPV = 48.63 ± 0.28
Max = 52.90 ± 0.30
χ²/ndf = 6.46
SNR = 2.40
r:1 c:2 m:1
Total events: 193481
Landau-Gauss fit, trigg.
L MPV = 47.20 ± 0.23
Max = 51.28 ± 0.25
χ²/ndf = 8.16
SNR = 2.97
r:1 c:3 m:1 layer 4
Total events: 193580
Landau-Gauss fit, trigg.
L MPV = 54.97 ± 0.40
Max = 59.92 ± 0.44
χ²/ndf = 4.34
SNR = 2.31
μ⁻-beam, E_b = 5 GeV
Run 464, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 193694
Landau-Gauss fit, trigg.
L MPV = 51.57 ± 0.33
Max = 55.71 ± 0.35
χ²/ndf = 5.61
SNR = 2.42
r:0 c:1 m:1
Total events: 194218
Landau-Gauss fit, trigg.
L MPV = 48.95 ± 0.27
Max = 53.17 ± 0.29
χ²/ndf = 6.92
SNR = 2.34
r:0 c:2 m:1
Total events: 193418
Landau-Gauss fit, trigg.
L MPV = 52.78 ± 0.36
Max = 57.61 ± 0.40
χ²/ndf = 5.96
SNR = 1.96
r:0 c:3 m:1
Total events: 129176
Landau-Gauss fit, trigg.
L MPV = 54.39 ± 0.52
Max = 59.10 ± 0.56
χ²/ndf = 2.66
SNR = 1.74
r:1 c:0 m:0
Total events: 194480
Landau-Gauss fit, trigg.
L MPV = 49.79 ± 0.35
Max = 53.64 ± 0.38
χ²/ndf = 5.09
SNR = 3.08
r:1 c:1 m:0
Total events: 194211
Landau-Gauss fit, trigg.
L MPV = 48.76 ± 0.27
Max = 52.96 ± 0.29
χ²/ndf = 6.72
SNR = 2.47
r:1 c:2 m:0
Total events: 194130
Landau-Gauss fit, trigg.
L MPV = 49.12 ± 0.26
Max = 53.31 ± 0.28
χ²/ndf = 5.70
SNR = 2.16
r:1 c:3 m:0
Total events: 194335
Landau-Gauss fit, trigg.
L MPV = 53.30 ± 0.43
Max = 58.13 ± 0.47
χ²/ndf = 4.24
SNR = 1.80
r:0 c:0 m:0
Total events: 194493
Landau-Gauss fit, trigg.
L MPV = 49.43 ± 0.28
Max = 52.83 ± 0.30
χ²/ndf = 4.06
SNR = 2.83
r:0 c:1 m:0
Total events: 194234
Landau-Gauss fit, trigg.
L MPV = 55.41 ± 0.39
Max = 60.31 ± 0.42
χ²/ndf = 5.98
SNR = 2.31
r:0 c:2 m:0
Total events: 194371
Landau-Gauss fit, trigg.
L MPV = 50.04 ± 0.33
Max = 54.23 ± 0.35
χ²/ndf = 5.10
SNR = 2.51
r:0 c:3 m:0
Total events: 194572
Landau-Gauss fit, trigg.
L MPV = 50.55 ± 0.51
Max = 54.02 ± 0.55
χ²/ndf = 1.91
SNR = 2.57
counts
ADC (arb. units)

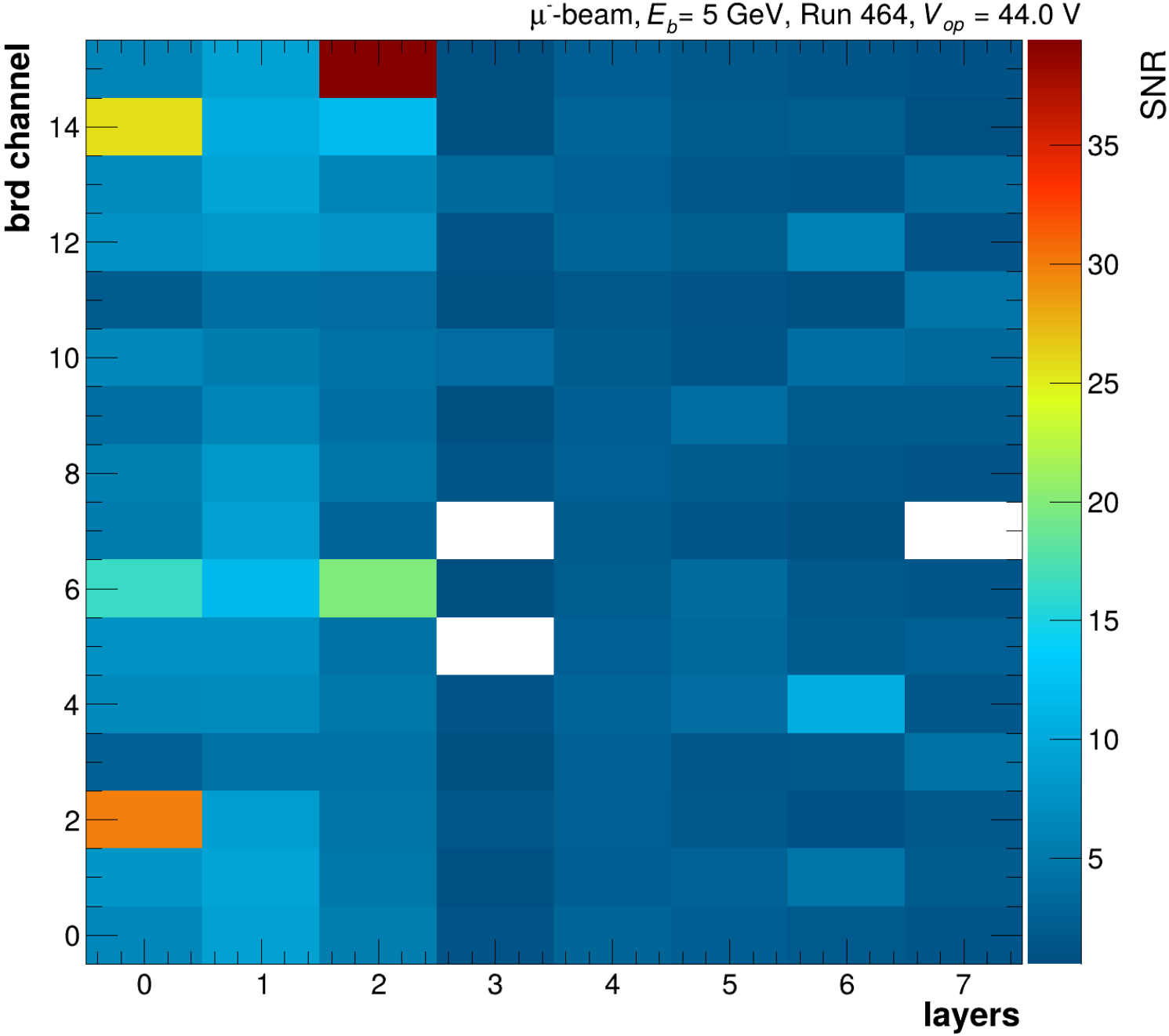

r:1 c:0 m:1
Total events: 193851
Landau-Gauss fit, trigg.
L MPV = 54.40 ± 0.30
Max = 58.18 ± 0.32
χ²/ndf = 5.15
SNR = 2.08
r:1 c:1 m:1
Total events: 193756
Landau-Gauss fit, trigg.
L MPV = 53.99 ± 0.24
Max = 57.63 ± 0.26
χ²/ndf = 4.90
SNR = 1.53
r:1 c:2 m:1
Total events: 112143
Landau-Gauss fit, trigg.
L MPV = 61.66 ± 0.27
Max = 65.84 ± 0.28
χ²/ndf = 6.38
SNR = 1.90
r:1 c:3 m:1 layer 5
Total events: 193757
Landau-Gauss fit, trigg.
L MPV = 57.43 ± 0.44
Max = 61.93 ± 0.47
χ²/ndf = 4.02
SNR = 1.71
μ⁻-beam, E_b = 5 GeV
Run 464, V_op = 44.0 V
r:0 c:0 m:1
Total events: 194557
Landau-Gauss fit, trigg.
L MPV = 57.33 ± 0.31
Max = 61.26 ± 0.33
χ²/ndf = 4.40
SNR = 1.93
r:0 c:1 m:1
Total events: 193734
Landau-Gauss fit, trigg.
L MPV = 53.64 ± 0.23
Max = 57.16 ± 0.24
χ²/ndf = 6.26
SNR = 3.77
r:0 c:2 m:1
Total events: 192970
Landau-Gauss fit, trigg.
L MPV = 58.49 ± 0.43
Max = 63.33 ± 0.47
χ²/ndf = 5.24
SNR = 1.42
r:0 c:3 m:1
Total events: 193846
Landau-Gauss fit, trigg.
L MPV = 56.15 ± 0.62
Max = 60.49 ± 0.67
χ²/ndf = 3.70
SNR = 1.19
r:1 c:0 m:0
Total events: 194549
Landau-Gauss fit, trigg.
L MPV = 57.98 ± 0.33
Max = 61.95 ± 0.36
χ²/ndf = 5.19
SNR = 3.68
r:1 c:1 m:0
Total events: 194400
Landau-Gauss fit, trigg.
L MPV = 60.81 ± 0.30
Max = 65.20 ± 0.32
χ²/ndf = 5.73
SNR = 3.22
r:1 c:2 m:0
Total events: 194444
Landau-Gauss fit, trigg.
L MPV = 52.11 ± 0.28
Max = 56.32 ± 0.31
χ²/ndf = 7.44
SNR = 3.55
r:1 c:3 m:0
Total events: 194539
Landau-Gauss fit, trigg.
L MPV = 56.94 ± 0.49
Max = 61.84 ± 0.53
χ²/ndf = 2.81
SNR = 1.33
r:0 c:0 m:0
Total events: 194584
Landau-Gauss fit, trigg.
L MPV = 60.71 ± 0.39
Max = 65.10 ± 0.42
χ²/ndf = 3.79
SNR = 2.24
r:0 c:1 m:0
Total events: 194513
Landau-Gauss fit, trigg.
L MPV = 54.97 ± 0.26
Max = 58.28 ± 0.27
χ²/ndf = 6.27
SNR = 2.81
r:0 c:2 m:0
Total events: 194492
Landau-Gauss fit, trigg.
L MPV = 59.44 ± 0.41
Max = 64.06 ± 0.44
χ²/ndf = 4.09
SNR = 1.74
r:0 c:3 m:0
Total events: 194627
Landau-Gauss fit, trigg.
L MPV = 57.09 ± 0.66
Max = 61.48 ± 0.72
χ²/ndf = 2.21
SNR = 1.50
counts
ADC (arb. units)

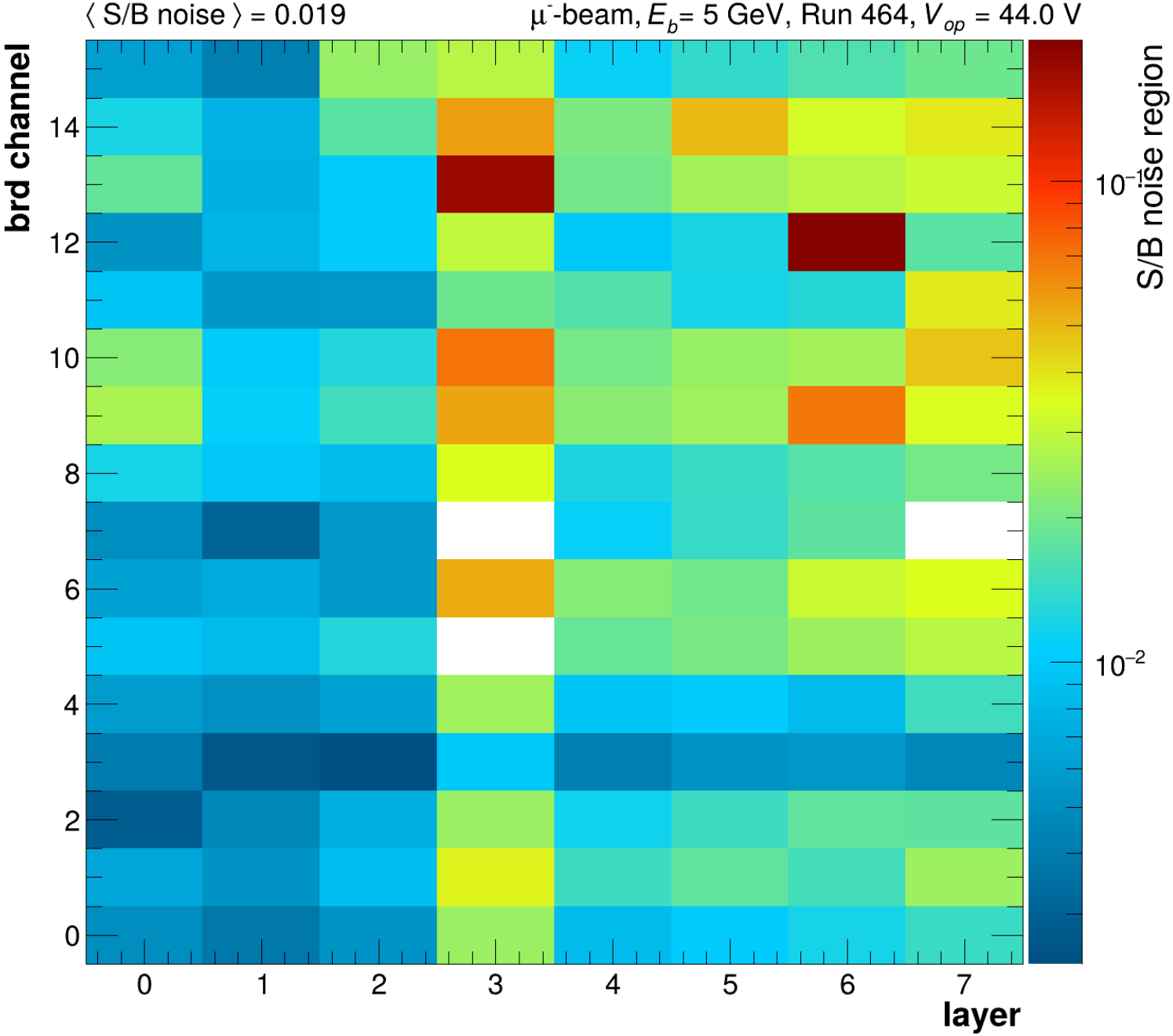

r:1 c:0 m:1
Total events: 20492
Landau-Gauss fit, trigg.
L MPV = 69.16 ± 0.47
Max = 74.26 ± 0.51
χ²/ndf = 7.10
SNR = 5.97
r:1 c:1 m:1
Total events: 193839
Landau-Gauss fit, trigg.
L MPV = 67.77 ± 0.37
Max = 72.74 ± 0.39
χ²/ndf = 5.09
SNR = 1.38
r:1 c:2 m:1
Total events: 193808
Landau-Gauss fit, trigg.
L MPV = 61.14 ± 0.28
Max = 65.28 ± 0.29
χ²/ndf = 7.88
SNR = 2.14
r:1 c:3 m:1 layer 6
Total events: 193897
Landau-Gauss fit, trigg.
L MPV = 68.84 ± 0.51
Max = 74.25 ± 0.55
χ²/ndf = 3.40
SNR = 1.40
μ⁻-beam, E_b = 5 GeV
Run 464, V_op = 44.0 V
r:0 c:0 m:1
Total events: 194604
Landau-Gauss fit, trigg.
L MPV = 65.02 ± 0.48
Max = 69.99 ± 0.51
χ²/ndf = 4.60
SNR = 1.54
r:0 c:1 m:1
Total events: 91221
Landau-Gauss fit, trigg.
L MPV = 72.53 ± 0.45
Max = 78.36 ± 0.49
χ²/ndf = 7.76
SNR = 1.81
r:0 c:2 m:1
Total events: 193795
Landau-Gauss fit, trigg.
L MPV = 66.07 ± 0.45
Max = 71.06 ± 0.48
χ²/ndf = 4.85
SNR = 3.98
r:0 c:3 m:1
Total events: 193882
Landau-Gauss fit, trigg.
L MPV = 69.42 ± 0.77
Max = 74.79 ± 0.83
χ²/ndf = 2.41
SNR = 0.90
r:1 c:0 m:0
Total events: 194601
Landau-Gauss fit, trigg.
L MPV = 61.80 ± 0.40
Max = 66.25 ± 0.43
χ²/ndf = 5.93
SNR = 10.41
r:1 c:1 m:0
Total events: 194526
Landau-Gauss fit, trigg.
L MPV = 64.81 ± 0.30
Max = 69.10 ± 0.31
χ²/ndf = 5.15
SNR = 1.95
r:1 c:2 m:0
Total events: 194551
Landau-Gauss fit, trigg.
L MPV = 63.27 ± 0.34
Max = 67.99 ± 0.36
χ²/ndf = 9.34
SNR = 1.75
r:1 c:3 m:0
Total events: 194587
Landau-Gauss fit, trigg.
L MPV = 60.57 ± 0.69
Max = 65.76 ± 0.75
χ²/ndf = 2.76
SNR = 0.82
r:0 c:0 m:0
Total events: 194636
Landau-Gauss fit, trigg.
L MPV = 66.41 ± 0.48
Max = 71.29 ± 0.52
χ²/ndf = 3.70
SNR = 1.78
r:0 c:1 m:0
Total events: 194610
Landau-Gauss fit, trigg.
L MPV = 59.74 ± 0.36
Max = 64.29 ± 0.39
χ²/ndf = 8.10
SNR = 4.55
r:0 c:2 m:0
Total events: 194604
Landau-Gauss fit, trigg.
L MPV = 60.19 ± 0.34
Max = 64.05 ± 0.36
χ²/ndf = 3.41
SNR = 1.10
r:0 c:3 m:0
Total events: 194644
Landau-Gauss fit, trigg.
L MPV = 60.70 ± 0.68
Max = 65.09 ± 0.73
χ²/ndf = 2.15
SNR = 1.74
counts
ADC (arb. units)

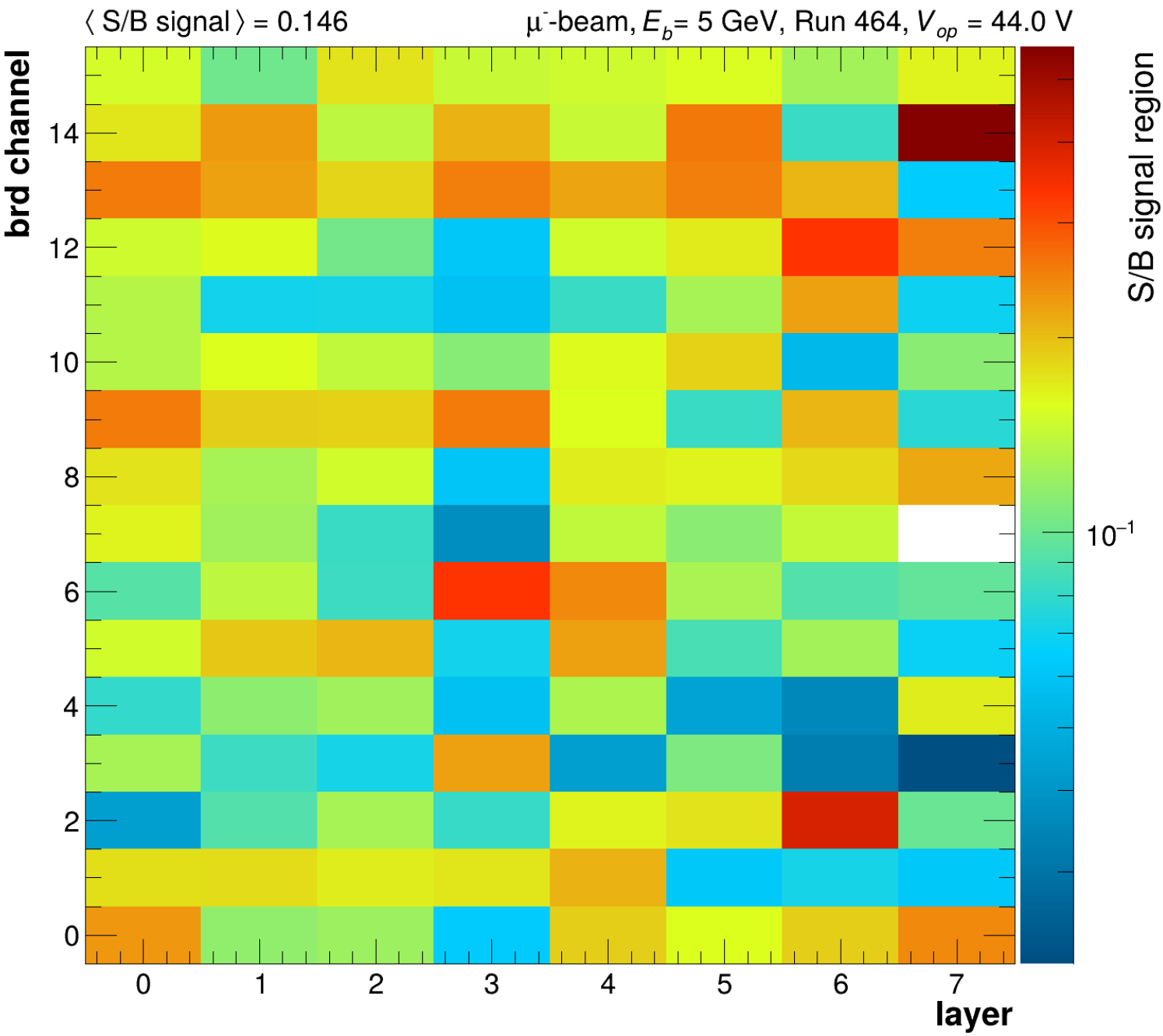

r:1 c:0 m:1
Total events: 194634
Landau-Gauss fit, trigg.
L MPV = 52.25 ± 0.40
Max = 57.01 ± 0.44
χ²/ndf = 2.95
SNR = 1.27
r:1 c:1 m:1
Total events: 193892
Landau-Gauss fit, trigg.
L MPV = 47.45 ± 0.25
Max = 50.92 ± 0.27
χ²/ndf = 4.94
SNR = 3.32
r:1 c:2 m:1
Total events: 193872
Landau-Gauss fit, trigg.
L MPV = 44.10 ± 0.30
Max = 48.52 ± 0.33
χ²/ndf = 7.97
SNR = 0.81
r:1 c:3 m:1 layer 7
Total events: 193094
Landau-Gauss fit, trigg.
L MPV = 48.56 ± 0.43
Max = 53.19 ± 0.47
χ²/ndf = 2.87
SNR = 1.06
μ⁻-beam, E_b = 5 GeV
Run 464, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 193937
Landau-Gauss fit, trigg.
L MPV = 51.43 ± 0.41
Max = 56.26 ± 0.44
χ²/ndf = 2.89
SNR = 1.19
r:0 c:1 m:1
Total events: 193869
Landau-Gauss fit, trigg.
L MPV = 49.66 ± 0.27
Max = 53.38 ± 0.29
χ²/ndf = 5.52
SNR = 1.92
r:0 c:2 m:1
Total events: 112244
Landau-Gauss fit, trigg.
L MPV = 49.06 ± 0.35
Max = 53.55 ± 0.39
χ²/ndf = 4.38
SNR = 3.14
r:0 c:3 m:1
Total events: 71571
Landau-Gauss fit, trigg.
L MPV = 53.79 ± 0.53
Max = 58.34 ± 0.57
χ²/ndf = 2.81
SNR = 4.44
r:1 c:0 m:0
Total events: 194646
Landau-Gauss fit, trigg.
L MPV = 47.19 ± 0.34
Max = 51.27 ± 0.37
χ²/ndf = 2.98
SNR = 1.55
r:1 c:1 m:0
Total events: 194608
Landau-Gauss fit, trigg.
L MPV = 49.90 ± 0.34
Max = 54.81 ± 0.38
χ²/ndf = 7.67
SNR = 2.52
r:1 c:2 m:0
Total events: 194609
Landau-Gauss fit, trigg.
L MPV = 50.79 ± 0.27
Max = 54.68 ± 0.29
χ²/ndf = 6.95
SNR = 1.42
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 194655
Landau-Gauss fit, trigg.
L MPV = 46.88 ± 0.36
Max = 51.10 ± 0.39
χ²/ndf = 2.85
SNR = 1.37
r:0 c:1 m:0
Total events: 194625
Landau-Gauss fit, trigg.
L MPV = 48.77 ± 0.30
Max = 52.25 ± 0.32
χ²/ndf = 3.76
SNR = 1.99
r:0 c:2 m:0
Total events: 194626
Landau-Gauss fit, trigg.
L MPV = 49.01 ± 0.35
Max = 53.02 ± 0.38
χ²/ndf = 5.15
SNR = 1.68
r:0 c:3 m:0
Total events: 194648
Landau-Gauss fit, trigg.
L MPV = 49.24 ± 0.48
Max = 52.35 ± 0.51
χ²/ndf = 1.98
SNR = 4.09
counts
ADC (arb. units)

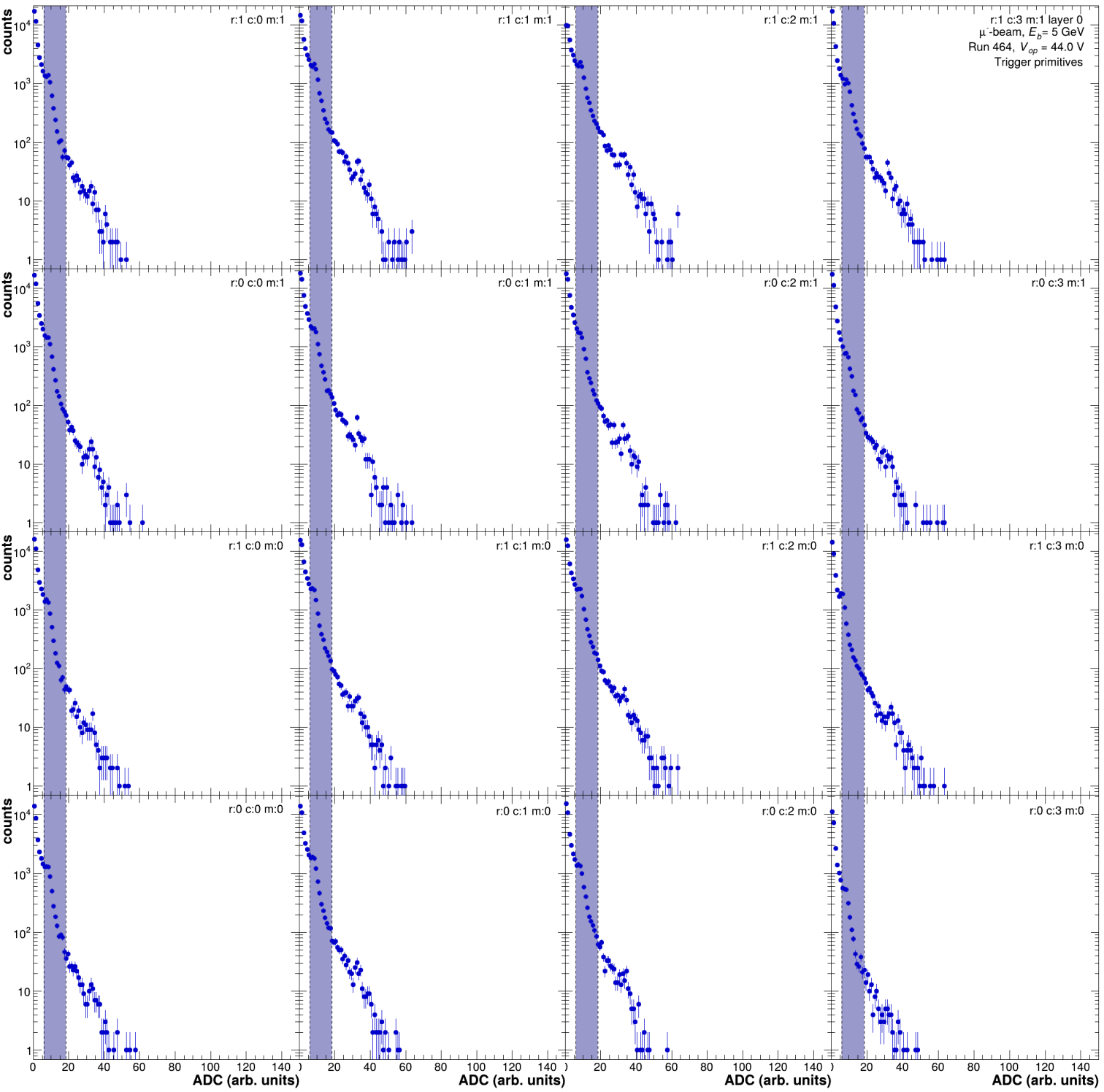

μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
100
90
80
70
60
50
40
30
20

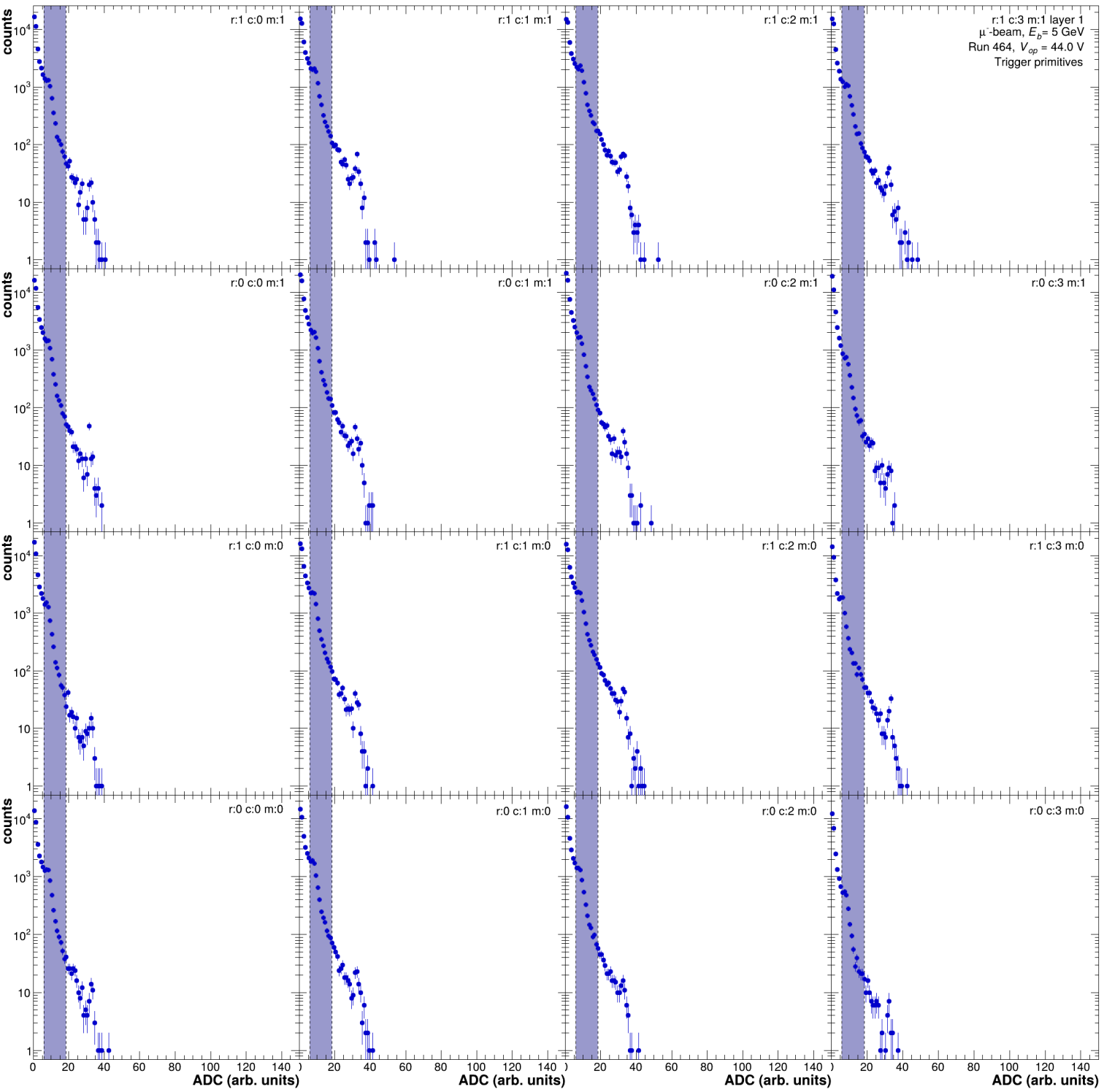

evt. collected = 194686
μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
layer
counts
10^4

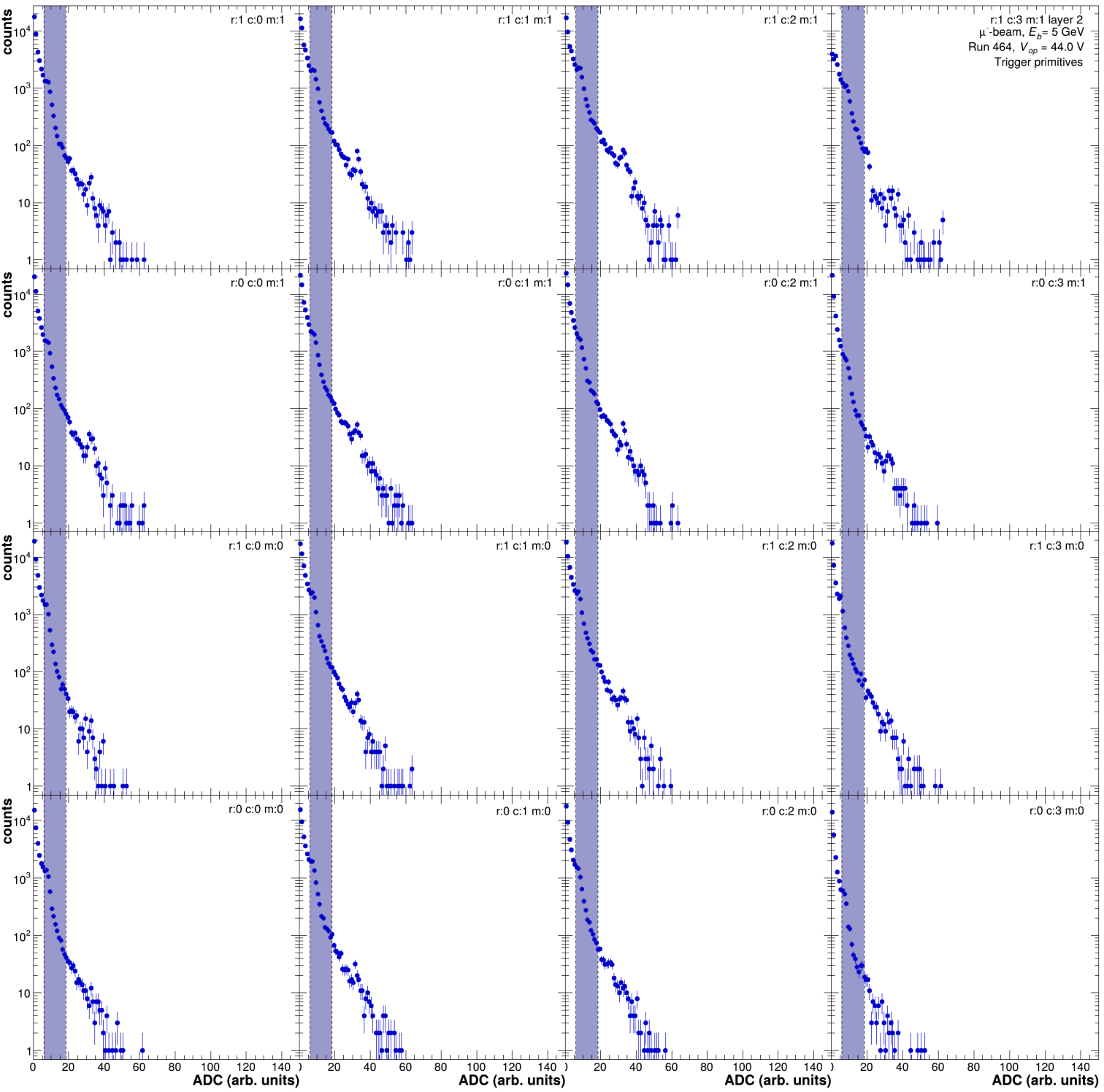

μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
5
10
15
20
25
30
35

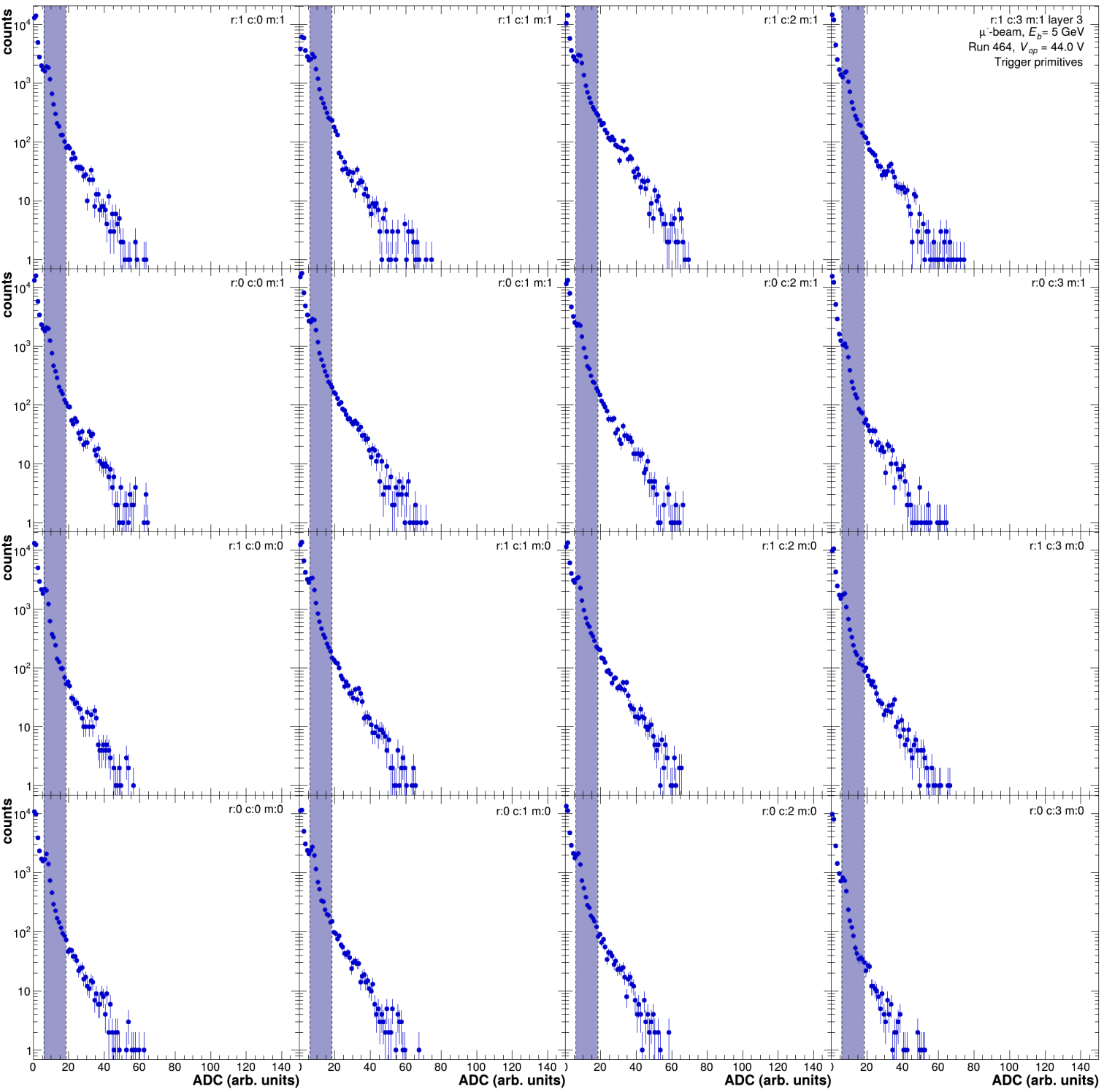

⟨ S/B noise ⟩ = 0.019
μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

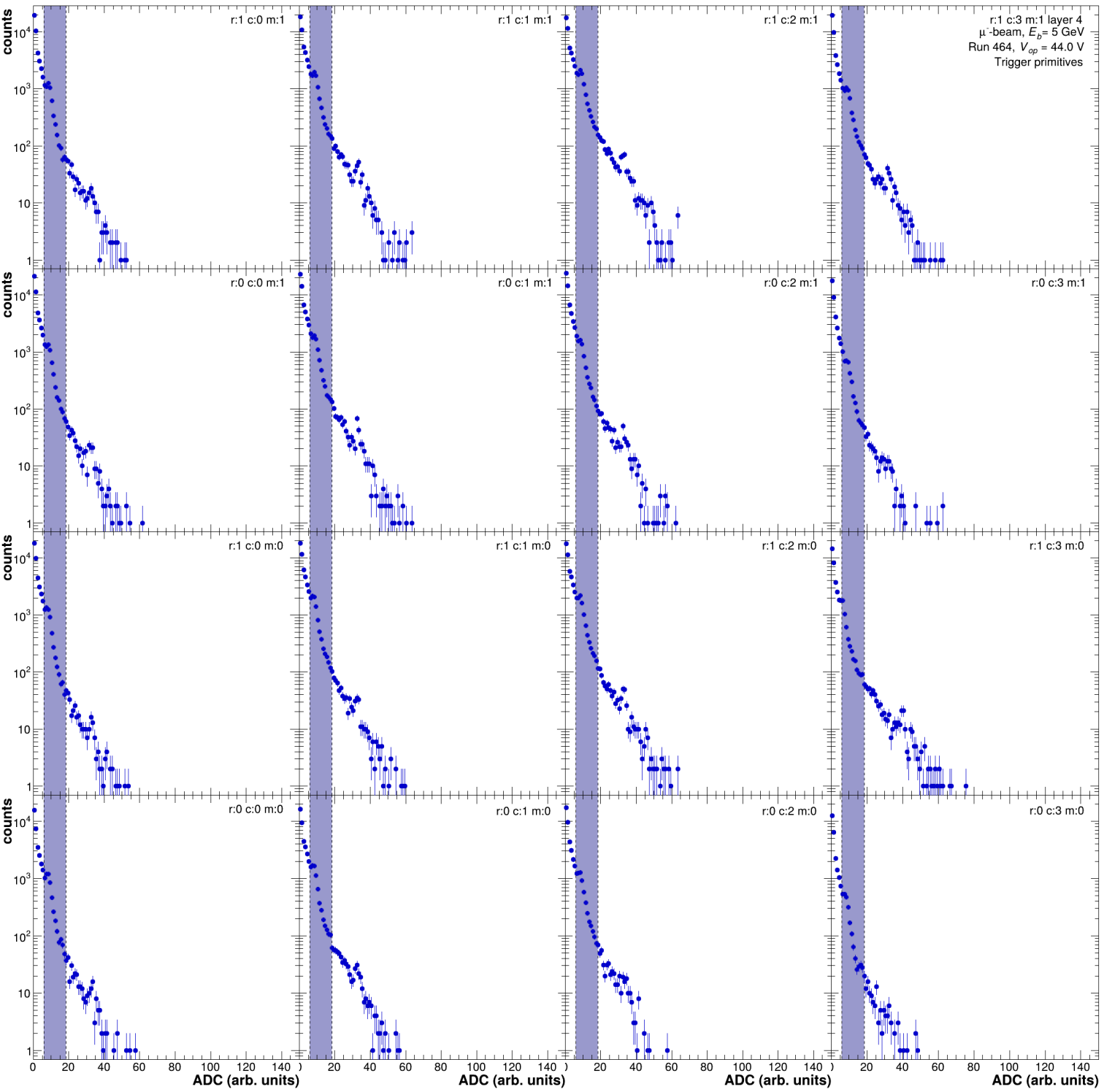

⟨ S/B signal ⟩ = 0.146
μ⁻-beam, E_b= 5 GeV, Run 464, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

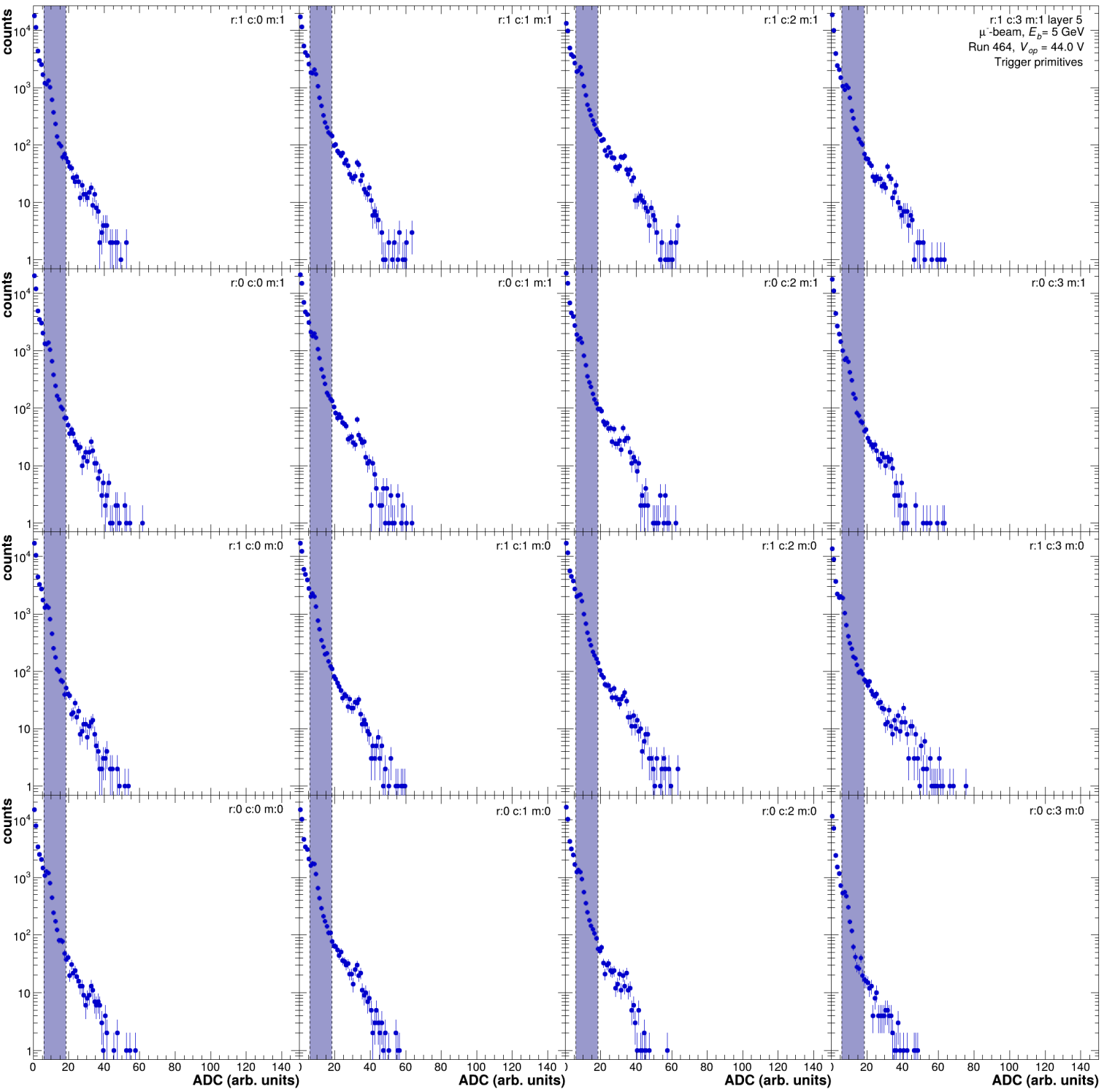

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

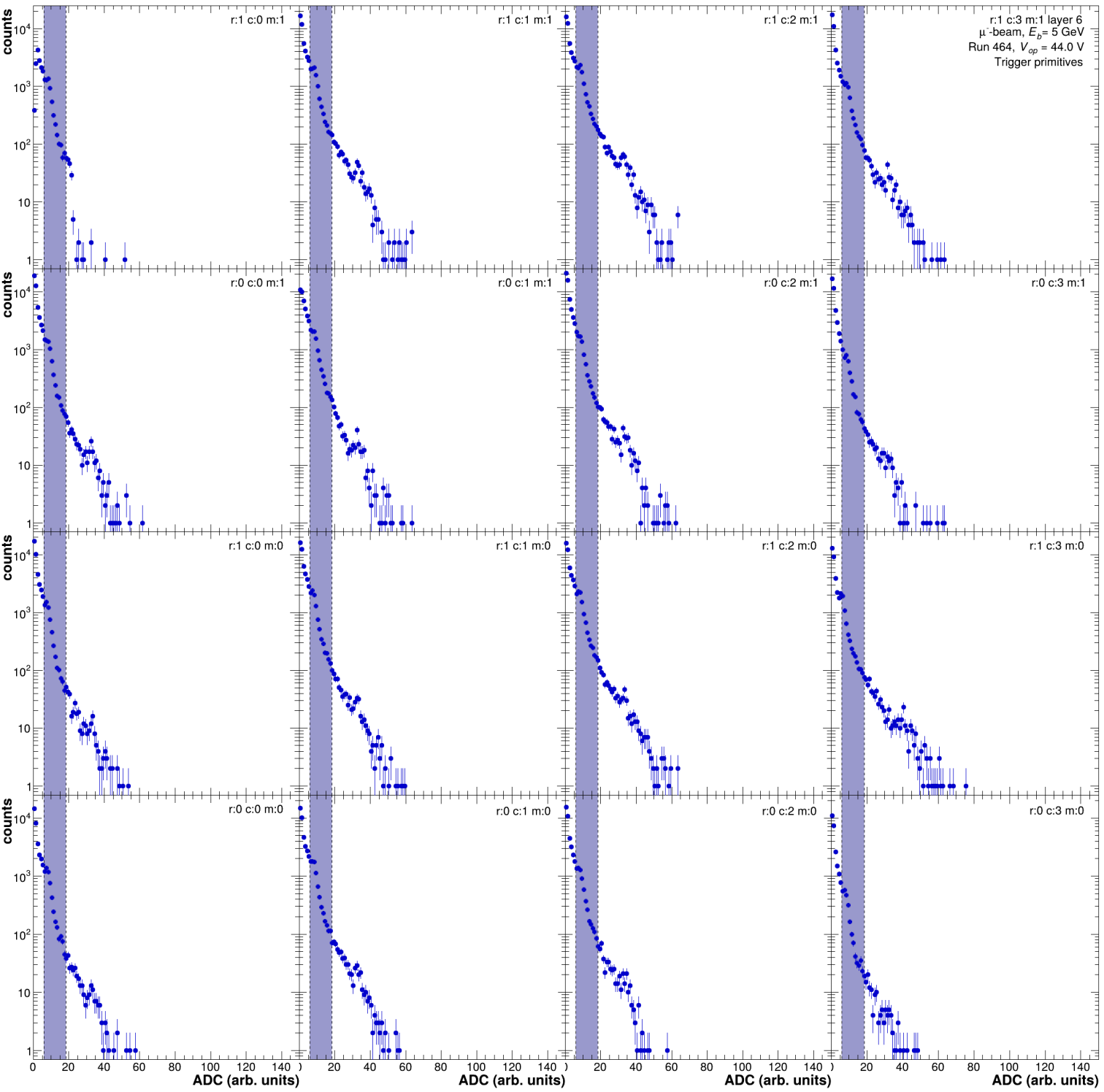

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

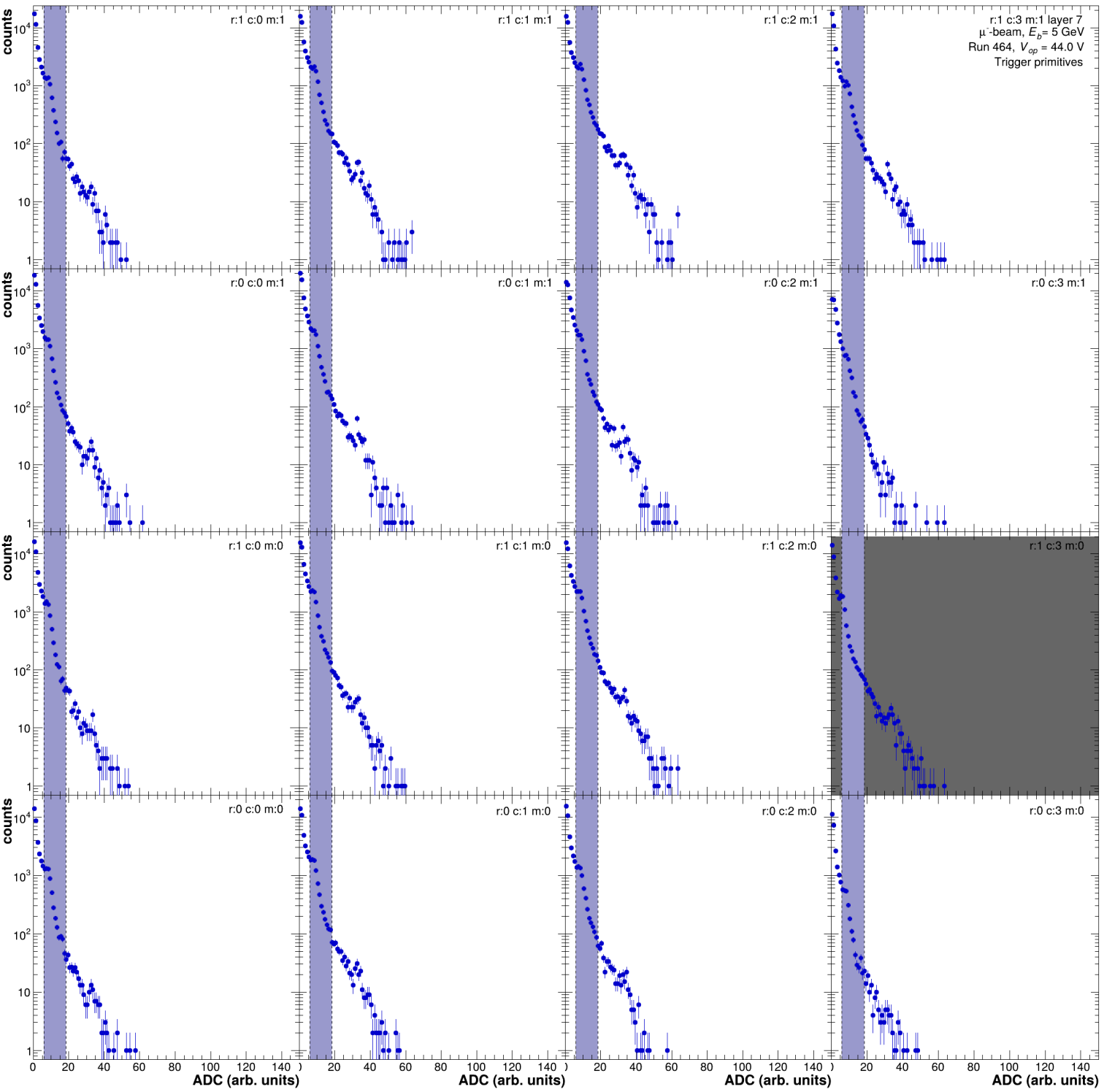

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 464, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)